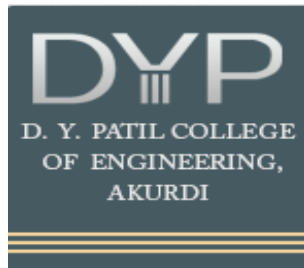


**D. Y.Patil College of Engineering Akurdi,
Pune -44**



Mini- Project

LAB MANUAL

Year: III

Semester: VI

**Prepared by:
Dr. B.A.Tingare
Mrs.D.A.Patil**



Mini Project (Cyber Security)

Laboratory

Course Code	Course Name	Teaching Scheme (Hrs./ Week)	Credits
317536	Mini Project(Cyber Security)	2	1

Course Objectives:

- To understand threats/vulnerabilities to networks and countermeasures.
- To provide understanding of cryptography and its applications.
- To explain various approaches to Encryption techniques.
- To understand working of firewall and IDS.

Course Outcomes:

On completion of the course, learner will be able to–

- CO1: Identify basic security attacks and services.
- CO2: Analyze the vulnerabilities and design a security solution.
- CO3: Implement symmetric and asymmetric key algorithms.
- CO4: Demonstrate network security applications, Firewall, IDS.

Operating System recommended: 64-bit Open source Linux or its derivative

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Sr. No	Title of Experiment (Cyber security)	CO Mapping	Page No
1	Implementation of S-DES	CO1	
2	Implementation of S-AES	CO1, CO2	
3	Implementation of Diffie-Hellman key exchange	CO3	
4	Implementation of RSA.	CO1, CO3	
5	Implementation of ECC algorithm	CO 4	
6	Enable/Configure (windows/Ubuntu) firewall. Create rules to filter network traffic and to block unauthorized network traffic.	CO4, CO5	
7	Configure and demonstrate an Intrusion Detection System (IDS) to detect suspicious activities and generate alerts when detected.	CO4, CO5	
Mini Project (any one)			
8	Mini Project 1: Implement Cross Site Scripting using stored attack. A stored cross-site scripting vulnerability in the comment functionality. [Note: To implement this assignment, submit a comment that calls the alert function when the blog post is viewed.]	CO4	
9	Mini Project 2: Implement SQL injection vulnerability attack that causes the application to display details of all the products available on website.	CO4	
10	Mini Project 3: Design the Access control vulnerability. [Note: This assignment has an unprotected admin panel. It is located at an unpredictable location, but the location is disclosed somewhere in the application. Use https://portswigger.net]	CO5	
11	Mini Project 4: This task is to demonstrate insecure and secured website. Develop a web site and demonstrate how the contents of the site can be changed by the attackers if it is http based and not secured. You can also add payment gateway and demonstrate how money transactions can be hacked by the hackers. Then support your website having https with SSL and demonstrate how secured website is.	CO5	

Lab Assignment No.	1
Title	Implementation of S-DES
Roll No.	
Class	TE
Date of Completion	
Subject	Mini Project(Cyber Security)
Assessment Marks	
Assessor's Sign	

ASSIGNMENT No: 01

Title: Implementation of S-DES (Data Encryption Standard)

Problem Statement: Implementation of S-DES

Prerequisite:

Basics of Computer networking and Python

Software Requirements:

Python 3

Hardware Requirements:

PIV, 2GB RAM, 500 GB HDD

Learning Objectives:

Learn Data Encryption Standard Algorithm (DES)

Outcomes:

After completion of this assignment students are able to understand the Data Encryption Standard.

Theory:

Data Encryption Standard (DES)

The Data Encryption Standard (DES) is a Symmetric-key block cipher issued by the national Institute of Standards & Technology (NIST).

DES is an implementation of a Feistel Cipher. It uses 16 round Feistel structure. The block size is 64-bit. Though, key length is 64-bit, DES has an effective key length of 56 bits, since 8 of the 64 bits of the key are not used by the encryption algorithm (function as check bits only). General Structure of DES is depicted in the following illustration –

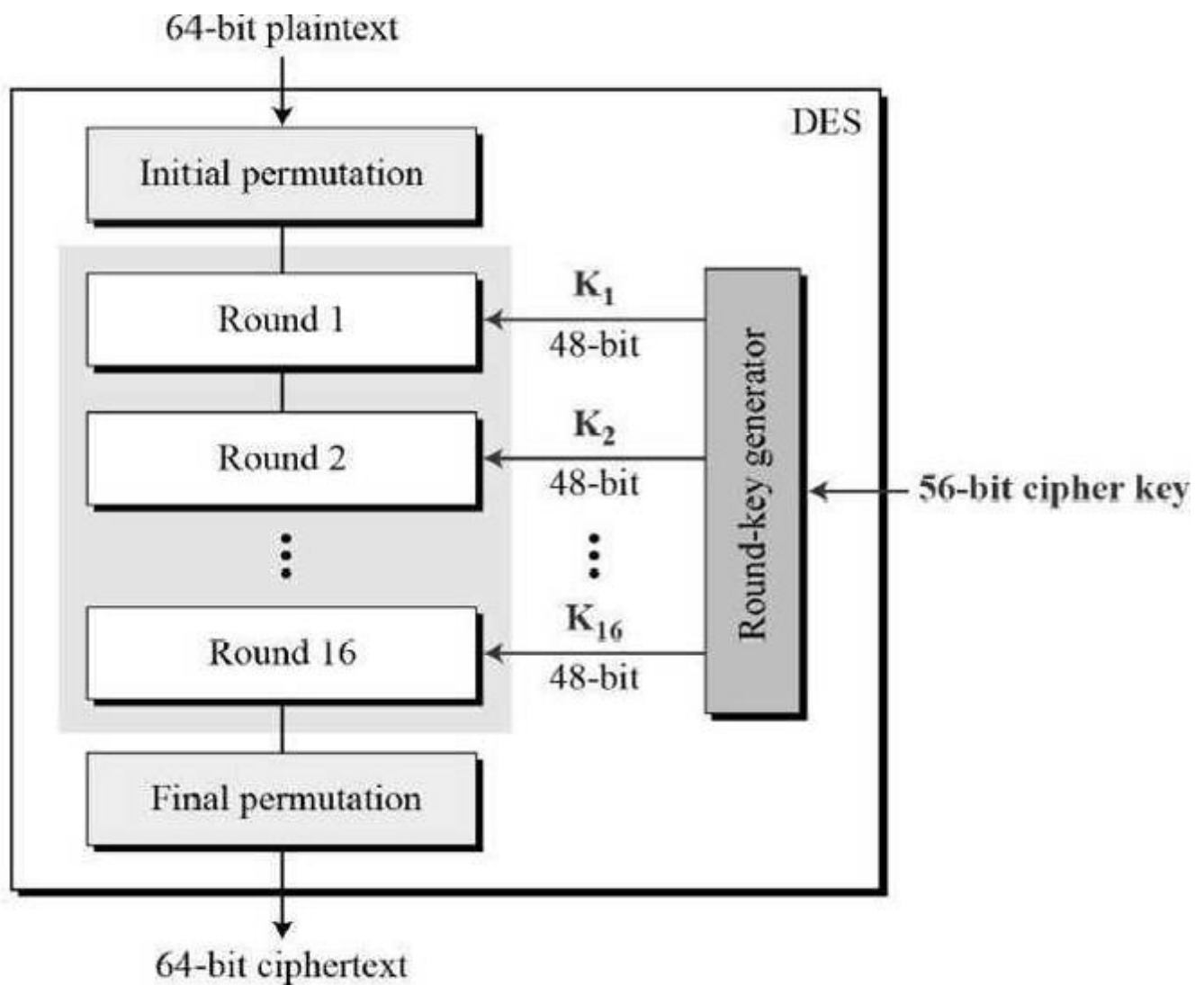


Figure 5.1: General Structure of DES

Since DES is based on the Feistel Cipher, all that is required to specify DES is –

- Round function
- Key schedule
- Any additional processing – Initial and final permutation

Initial and Final Permutation

The initial and final permutations are straight Permutation boxes (P-boxes) that are inverses of each other. They have no cryptography significance in DES. The initial and final permutations are shown as follows

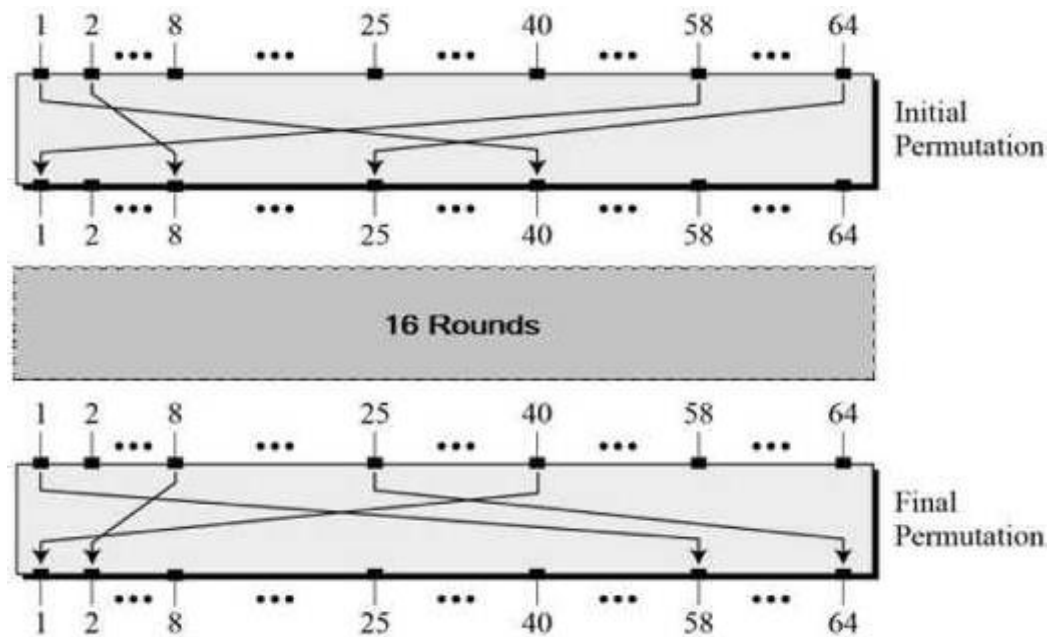


Figure 5.2 initial and final permutations

Round Function

The heart of this cipher is the DES function, f . The DES function applies a 48-bit key to the rightmost 32 bits to produce a 32-bit output.

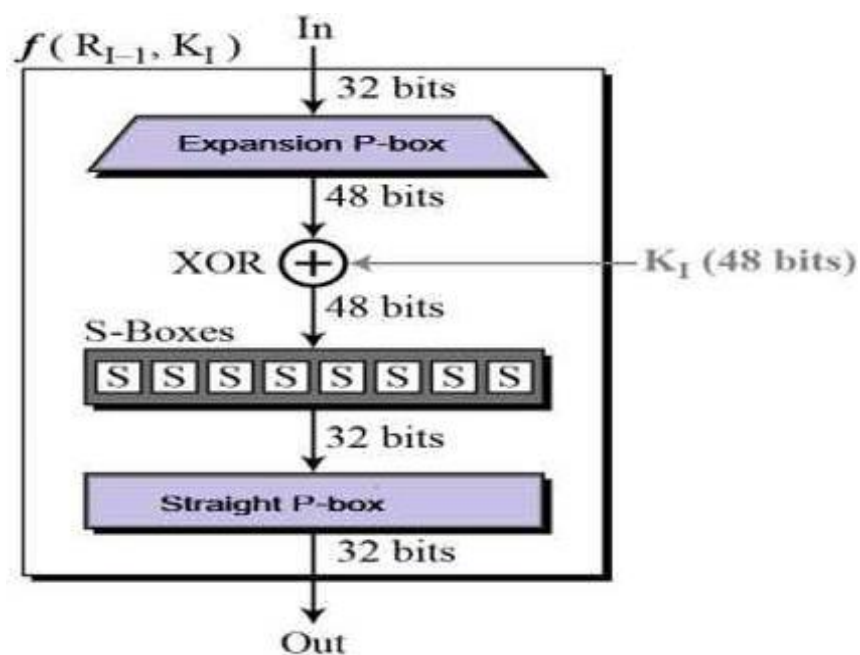
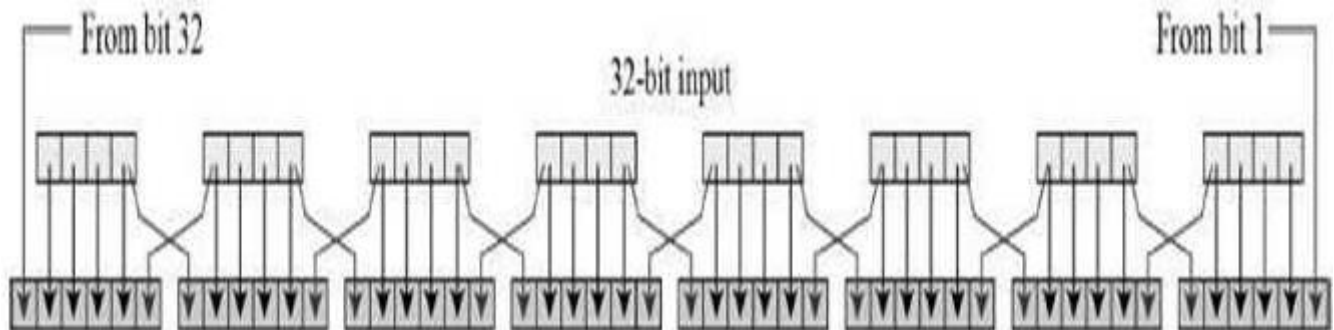


Figure 5.3 Round Functions

Expansion Permutation Box

Since right input is 32-bit and round key is a 48-bit, we first need to expand right input to 48 bits.



Permutation logic is graphically depicted in the following illustration:

Figure 5.4 Permutation logic

The graphically depicted Permutation logic is generally described as table in DES specification illustrated as shown:

Table 5.1 Permutation logic

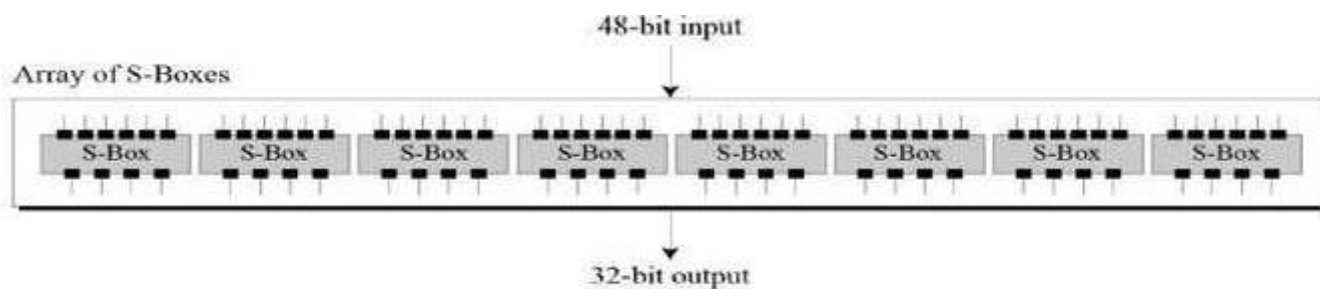
32	01	02	03	04	05
04	05	06	07	08	09
08	09	10	11	12	13
12	13	14	15	16	17
16	17	18	19	20	21
20	21	22	23	24	25
24	25	26	27	28	29
28	29	31	31	32	01

XOR(Whitener)

After the expansion permutation, DES does XOR operation on the expanded right section and the round key. The round key is used only in this operation.

Substitution Boxes

The S-boxes carry out the real mixing (confusion). DES uses 8 S-boxes, each with a 6-bit input and



a 4-bit output. Refer the following illustration –

Figure 5.5 S-Boxes

The S-box rule is illustrated below –

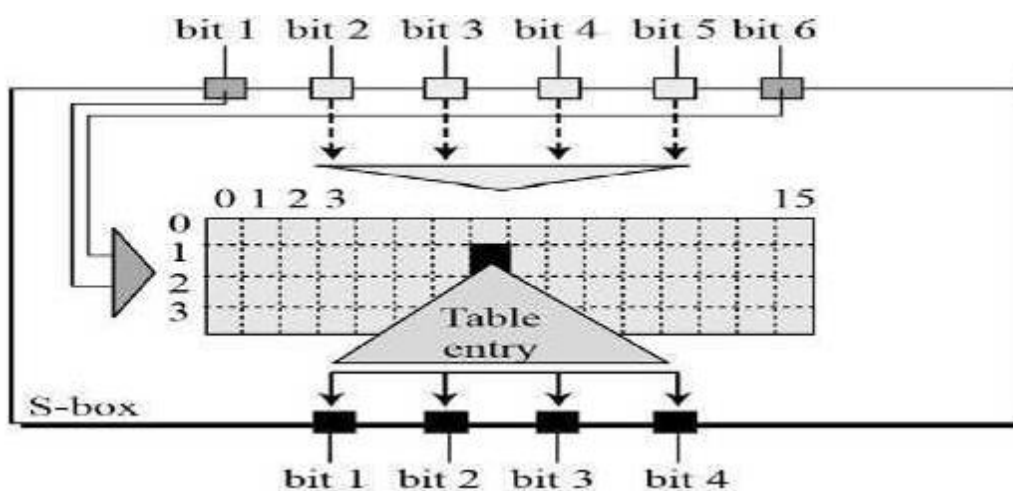


Figure 5.6 S-Box Rules

There are a total of eight S-box tables. The output of all eight s-boxes is then combined in to 32 bit section.

Straight Permutation – The 32 bit output of S-boxes is then subjected to the straight permutation with rule shown in the following illustration:

Table 5.2 Straight Permutation

16	07	20	21	29	12	28	17
01	15	23	26	05	18	31	10
02	08	24	14	32	27	03	09
19	13	30	06	22	11	04	25

Key Generation

The round-key generator creates sixteen 48-bit keys out of a 56-bit cipher key. The process of key generation is depicted in the following illustration –

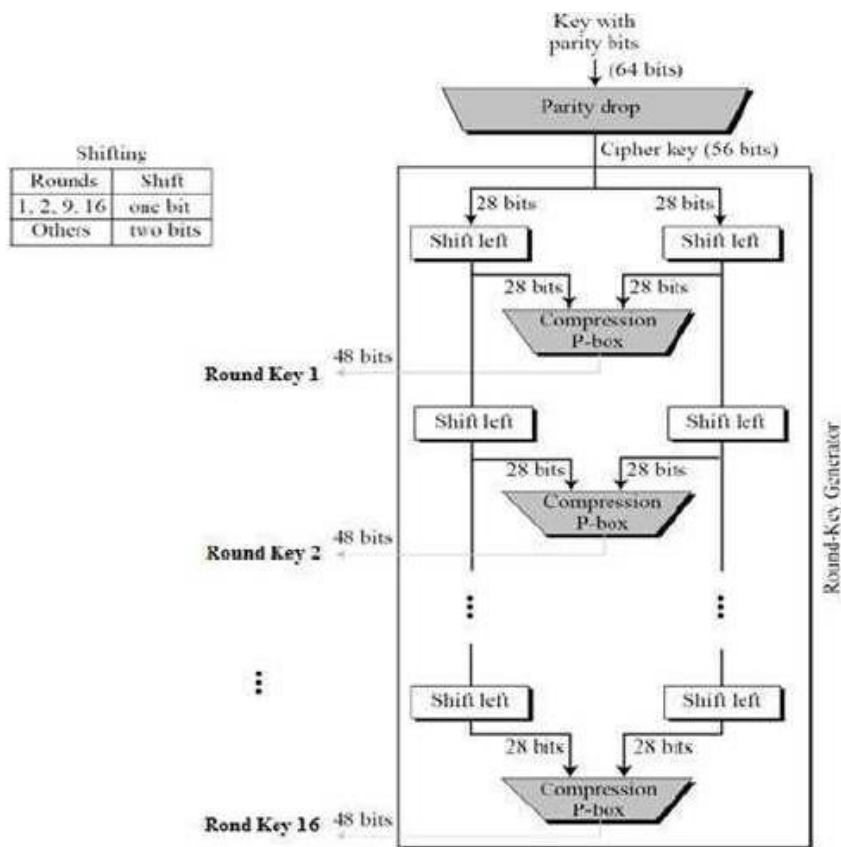


Figure 5.7 the process of key generation

The logic for Parity drops, shifting, and Compression P-box is given in the DES description.

DES Analysis

The DES satisfies both the desired properties of block cipher. These two properties make cipher very strong.

- **Avalanche effect** – A small change in plaintext results in the very great change in the cipher text.
- **Completeness** – Each bit of cipher text depends on many bits of plaintext.

During the last few years, cryptanalysis has found some weaknesses in DES when key selected are weak keys. These keys shall be avoided.

DES has proved to be a very well designed block cipher. There have been no significant cryptanalytic attacks on DES other than exhaustive key search.

Conclusion:

Thus we have successfully studied implementation of S-DES

Oral Questions :

1. Explain structure of DES
2. Explain Initial permutation steps in DES
3. Draw and explain the internal structure of single round of DES
4. Explain Double DES
5. Explain Triple DES

Lab Assignment No.	2
Title	Implementation of S-AES
Roll No.	
Class	TE
Date of Completion	
Subject	Mini Project(Cyber Security)
Assessment Marks	
Assessor's Sign	

ASSIGNMENT No: 02

Title: Implementation of S-AES (Advanced Encryption Standard)

Problem Statement: Implementation of S-AES

Prerequisite:

Basic of Python3, Concept of Advanced Encryption Standard

Software Requirements:

Python 3

Hardware Requirement:

PIV, 2GB RAM, 500 GB HDD

Learning Objectives:

Learn How to Apply Advanced Encryption Standard Algorithm to encryption of given data.

Outcomes:

After completion of this assignment students are able Implement code for **Advanced Encryption Standard Algorithm** for given data and find the encrypted data of the given data.

Theory: The more popular and widely adopted symmetric encryption algorithm likely to be encountered nowadays is the Advanced Encryption Standard (AES). It is found at least six times faster than triple DES.

A replacement for DES was needed as its key size was too small. With increasing computing power, it was considered vulnerable against exhaustive key search attack. Triple DES was designed to overcome this drawback but it was found slow.

The features of AES are as follows –

- Symmetric key symmetric block cipher
- 128-bit data, 128/192/256-bit keys
- Stronger and faster than Triple-DES
- Provide full specification and design details

- Software implementable in C ,Java and Python

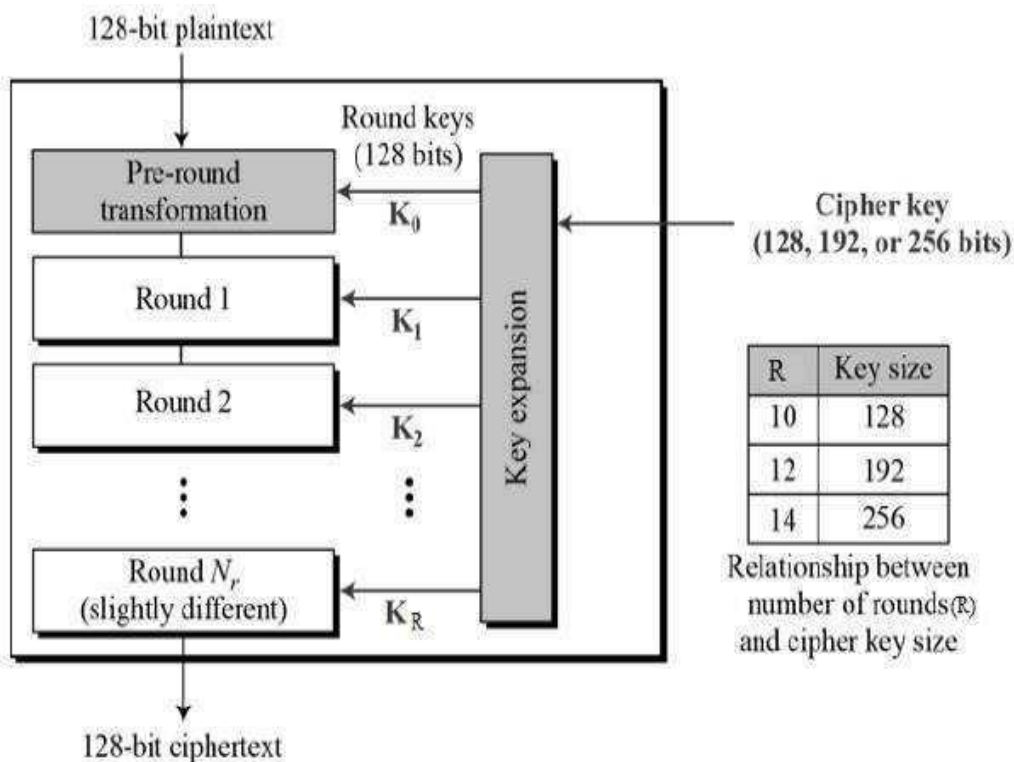
Operation of AES

AES is an iterative rather than Feistel cipher. It is based on ‘substitution–permutation network’. It comprises of a series of linked operations, some of which involve replacing inputs by specific outputs (substitutions) and others involve shuffling bits around (permutations).

Interestingly, AES performs all its computations on bytes rather than bits. Hence, AES treats the 128 bits of a plaintext block as 16 bytes. These 16 bytes are arranged in four columns and four rows for processing as a matrix –

Unlike DES, the number of rounds in AES is variable and depends on the length of the key. AES uses 10 rounds for 128-bit keys, 12 rounds for 192-bit keys and 14 rounds for 256-bit keys. Each of these rounds uses a different 128-bit round key, which is calculated from the original AES key.

The schematic of AES structure is given in the following illustration –



Encryption Process

Here, we restrict to description of a typical round of AES encryption. Each round comprise of four sub-processes. The first round process is depicted below –

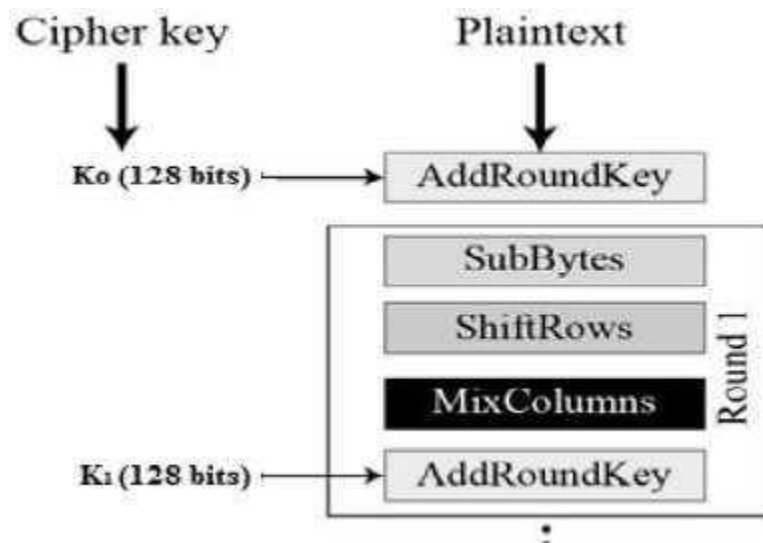


Figure 6.2 Encryption Process

Conclusion:

Thus we have studied implementation of S-AES

Oral Questions :

1. Explain Working of AES algorithm in details
2. What are the features of AES as Compared with DES
3. Explain the operation in key Expansion process in AES
4. Explain operation of AES algorithm and state its application

Lab Assignment No.	03
Title	Implementation of Diffie-Hellman key exchange
Roll No.	
Class	TE
Date of Completion	
Subject	Mini Project(Cyber Security)
Assessment Marks	
Assessor's Sign	

ASSIGNMENT No: 03

Title: Implementation of Diffie-Hellman key Exchange (DH)

Problem Statement: Implementation of Diffie-Hellman key Exchange (DH)

Prerequisite:

Basics of Computer Networking and Python

Software Requirements:

Python 3

Hardware Requirements:

PIV, 2GB RAM, 500 GB HDD

Learning Objectives:

Learn Diffie-Hellman key Exchange (DH)

Outcomes:

After completion of this assignment students are able to understand the Diffie-Hellman key Exchange

Theory Concepts:

Diffie-Hellman key Exchange (DH)

In the mid- 1970's, Whitefield Diffie, a student at the Stanford University met with Martin Hellman, his professor & the two began to think about it. After some research & complicated mathematical analysis, they came up with the idea of AKC. Many experts believe that this development is the first & perhaps the only truly revolutionary concept in the history of cryptography.

Silent Features of Diffie-Hellman key Exchange (DH)

1. Developed to address shortfalls of *key distribution* in symmetric key distribution.
2. A *key exchange algorithm*, not an encryption algorithm
3. Allows two users to share a *secret key* securely over a public network
4. Once the key has been shared Then both parties can use it to encrypt and decrypt messages using symmetric cryptography
5. Algorithm is based on “difficulty of calculating discrete logarithms in a finite field”
6. These keys are mathematically related to each other.
7. “Using the public key of users, the session key is generated without transmitting the private key of the users.”

Diffie-Hellman Key Exchange/Agreement Algorithm with Example

1. Firstly, Alice and Bob agree on two large prime numbers, n and g . These two integers need not be kept secret. Alice and Bob can use an insecure channel to agree on them.

Let $n = 11$, $g = 7$.
2. Alice chooses another large random number x , and calculates A such that:
 $A = g^x \bmod n$

Let $x = 3$. Then, we have, $A = 7^3 \bmod 11 = 343 \bmod 11 = 2$.
3. Alice sends the number A to Bob.

Alice sends 2 to Bob.
4. Bob independently chooses another large random integer y and calculates B such that:
 $B = g^y \bmod n$

Let $y = 6$. Then, we have, $B = 7^6 \bmod 11 = 117649 \bmod 11 = 4$.
5. Bob sends the number B to Alice.

Bob sends 4 to Alice.
6. A now computes the secret key $K1$ as follows:
 $K1 = B^x \bmod n$

We have, $K1 = 4^3 \bmod 11 = 64 \bmod 11 = 9$.
7. B now computes the secret key $K2$ as follows:
 $K2 = A^y \bmod n$

We have, $K2 = 2^6 \bmod 11 = 64 \bmod 11 = 9$.

Diffie-Hellman Key exchange

1. Public values:
 - large prime p , generator g (primitive root of p)
2. Alice has secret value x , Bob has secret y
 Discrete logarithm problem: given x , g , and n , find A
3. $A \equiv B: g^x \pmod{n}$
4. $B \equiv A: g^y \pmod{n}$
5. Bob computes $(g^x)^y = g^{xy} \pmod{n}$
6. Alice computes $(g^y)^x = g^{xy} \pmod{n}$
7. Symmetric key = $g^{xy} \pmod{n}$

Limitation: Vulnerable to “man in the middle” attacks*

Man-in-the-Middle Attack:

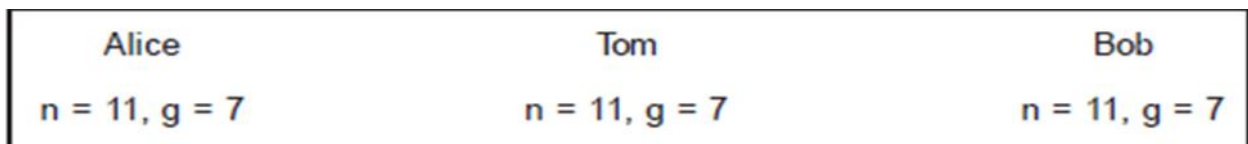


Figure 7.1 Man-in-the-Middle Attack Part-I

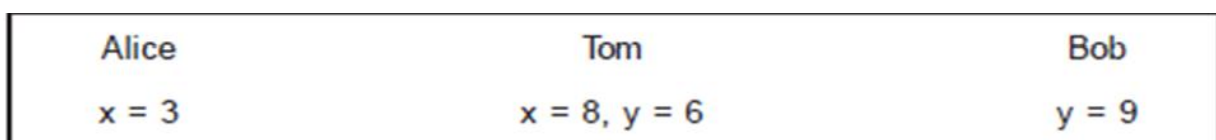


Figure 7.2 Man-in-the-Middle Attack Part-II

Alice	Tom	Bob
$A = g^x \bmod n$ $= 7^3 \bmod 11$ $= 343 \bmod 11$ $= 2$	$A = g^x \bmod n$ $= 7^3 \bmod 11$ $= 5764801 \bmod 11$ $= 9$ $B = g^y \bmod n$ $= 7^6 \bmod 11$ $= 117649 \bmod 11$ $= 4$	$B = g^y \bmod n$ $= 7^9 \bmod 11$ $= 40353607 \bmod 11$ $= 8$

Figure 7.3 Man-in-the-Middle Attack Part-III

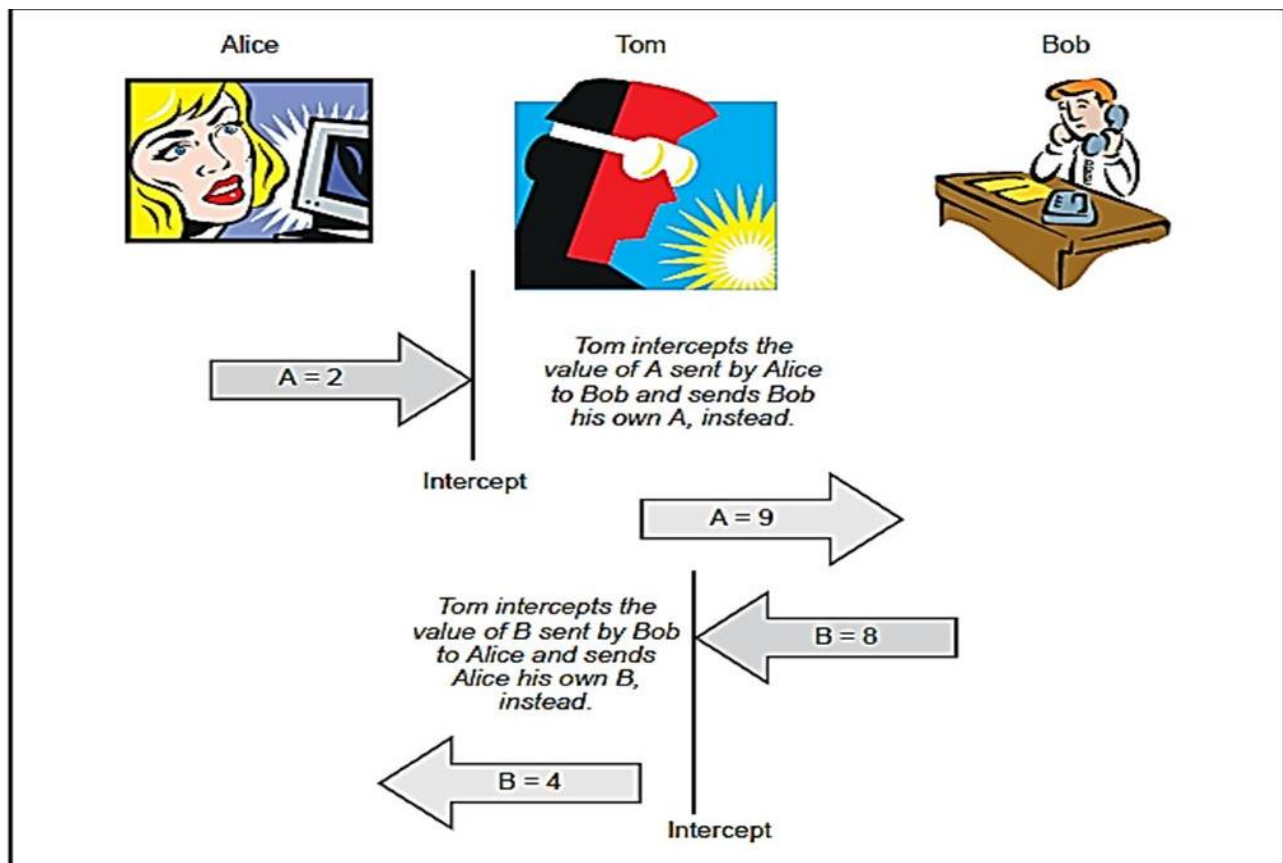


Figure 7.4 Man-in-the-Middle Attack Part-IV

Alice	Tom	Bob
$A = 2, B = 4^*$	$A = 2, B = 8$	$A = 9^*, B = 8$

(Note: * indicates that these are the values after Tom hijacked and changed them.)

Figure 7.5 Man-in-the-Middle Attack Part-V

Alice	Tom	Bob
$K1 = B^x \bmod n$	$K1 = B^x \bmod n$	$K2 = A^y \bmod n$
$= 4^3 \bmod 11$	$= 8^8 \bmod 11$	$= 9^9 \bmod 11$
$= 64 \bmod 11$	$= 16777216 \bmod 11$	$= 387420489 \bmod 11$
$= 9$	$= 5$	$= 5$
	$K2 = A^y \bmod n$	
	$= 2^8 \bmod 11$	
	$= 64 \bmod 11$	
	$= 9$	

Figure 7.6 Man-in-the-Middle Attack Part-VI

Preventing a Man-in-the-Middle Attack with Hashing

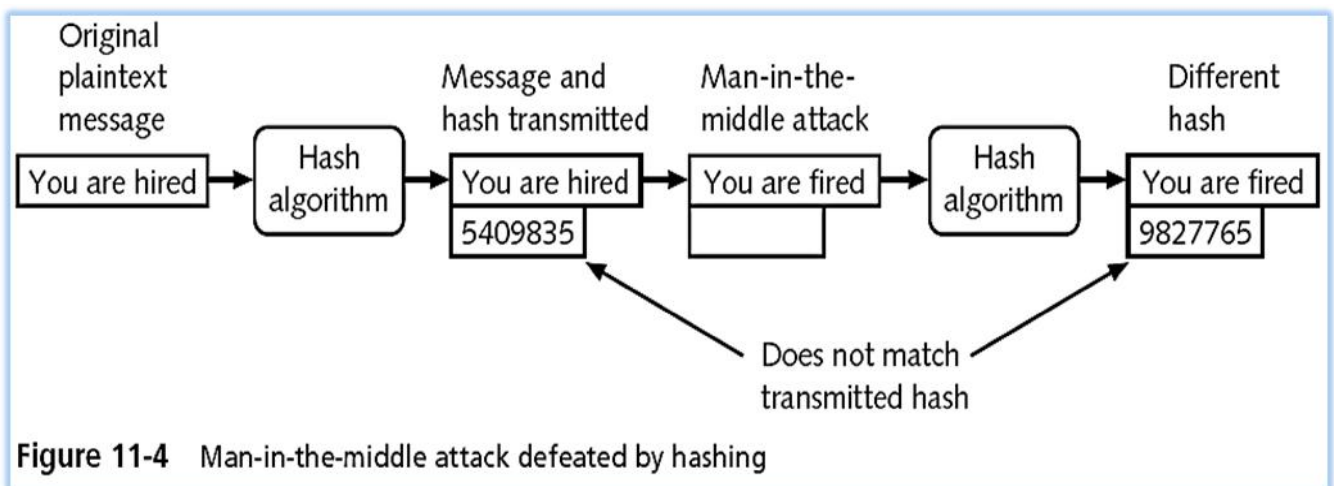


Figure 11-4 Man-in-the-middle attack defeated by hashing

Conclusion:

Thus we have studied and implement Diffie-Hellmen key exchange algorithm and how to prevent Man-in-the-Middle Attack

Oral Questions :

1. Explain "Diffie-Hellmen key exchange algorithm with suitable example
2. What is Man in the middle attack?
3. How to Preventing a Man-in Middle Attack?

Lab Assignment No.	04
Title	Implementation of RSA.
Roll No.	
Class	TE
Date of Completion	
Subject	Mini Project(Cyber Security)
Assessment Marks	
Assessor's Sign	

ASSIGNMENT No: 04

Title: Implementation of RSA.

Problem Statement: Implementation of RSA Algorithm

Prerequisite:

Basic of Python, Information security Algorithm

Software Requirements:

Python 3.7

Hardware Requirement:

2GB RAM, 500 GB HDD

Learning Objectives:

Learn RSA Algorithm

Outcomes:

After completion of this assignment students are able to understand the How asymmetric cryptographic algorithm is used

Theory Concepts:

RSA(Rivest, Shamir & Adleman)

RSA is an algorithm used by modern computers to encrypt and decrypt messages. It is an asymmetric cryptographic algorithm. Asymmetric means that there are two different keys. This is also called public key cryptography, because one of the keys can be given to anyone. The other key must be kept private. The algorithm is based on the fact that finding the factors of a large composite number is difficult: when the integers are prime numbers, the problem is called prime factorization. It is also a key pair (public and private key) generator.

- RSA makes the public and private keys by multiplying two large prime numbers p and q
 - It's easy to find & multiply large prime No. ($n=pq$)
 - It is very difficult to factor the number n to find p and q
 - Finding the private key from the public key would require a factoring operation
 - The real challenge is the selection & generation of keys.
- RSA is complex and slow, but secure
- 100 times slower than DES on s/w & 1000 times on h/w

The Rivest-Shamir-Adleman (RSA) algorithm is one of the most popular and secures public- key encryption methods. The algorithm capitalizes on the fact that there is no efficient way to factor very large (100-200 digit) numbers.

Using an encryption key (e,n) , the algorithm is as follows:

1. Represent the message as an integer between 0 and $(n-1)$. Large messages can be broken up into a number of blocks. Each block would then be represented by an integer in the same range.
2. Encrypt the message by raising it to the e th power modulo n . The result is a ciphertext message C .
3. To decrypt ciphertext message C , raise it to another power d modulo n

The encryption key (e,n) is made public. The decryption key (d,n) is kept private by the user.

How to Determine Appropriate Values for e , d , and n

1. Choose two very large (100+ digit) prime numbers. Denote these numbers as p and q .
2. Set n equal to $p * q$.
3. Choose any large integer, d , such that $\text{GCD}(d, ((p-1) * (q-1))) = 1$
4. Find e such that $e * d = 1 \pmod{((p-1) * (q-1))}$

Rivest, Shamir, and Adleman provide efficient algorithms for each required operation[4].

How secure is a communication using RSA?

Cryptographic methods cannot be proven secure. Instead, the only test is to see if someone can figure out how to decipher a message without having direct knowledge of the decryption key. The RSA method's security rests on the fact that it is extremely difficult to factor very large numbers. If 100 digit numbers are used for p and q , the resulting n will be approximately 200 digits. The fastest known factoring algorithm would take far too long for an attacker to ever break the code. Other methods for determining d without factoring n are equally as difficult.

Any cryptographic technique which can resist a concerted attack is regarded as secure. At this point in time, the RSA algorithm is considered secure.

How Does RSA Works?

RSA is an asymmetric system, which means that a key pair will be generated (we will see how soon) , a public key and a private key , obviously you keep your private key secure and pass around the public one.

The algorithm was published in the 70's by Ron Rivest, Adi Shamir, and Leonard Adleman, hence RSA, and it sort of implement's a trapdoor function such as Diffie's one.

RSA is rather slow so it's hardly used to encrypt data, more frequently it is used to encrypt and pass around symmetric keys which can actually deal with encryption at a faster speed.

RSA Security:

- It uses prime number theory which makes it difficult to find out the key by reverse engineering.
- Mathematical Research suggests that it would take more than 70 years to find P & Q if N is a 100 digit number.

Algorithm

The RSA algorithm holds the following features –

- RSA algorithm is a popular exponentiation in a finite field over integers including prime numbers.
- The integers used by this method are sufficiently large making it difficult to solve.
- There are two sets of keys in this algorithm: private key and public key.

You will have to go through the following steps to work on RSA algorithm –

Step 1: Generate the RSA modulus

The initial procedure begins with selection of two prime numbers namely p and q , and then calculating their product N , as shown –

$$N = p * q$$

Here, let N be the specified large number.

Step 2: Derived Number (e)

Consider number e as a derived number which should be greater than 1 and less than $(p-1)$ and $(q-1)$. The primary condition will be that there should be no common factor of $(p-1)$ and $(q-1)$ except 1

Step 3: Public key

The specified pair of numbers n and e forms the RSA public key and it is made public.

Step 4: Private Key

Private Key d is calculated from the numbers p , q and e . The mathematical relationship between the numbers is as follows –

$$ed = 1 \text{ mod } (p-1)(q-1)$$

The above formula is the basic formula for Extended Euclidean Algorithm, which takes p and q as the input parameters.

Encryption Formula

Consider a sender who sends the plain text message to someone whose public key is (n,e) . To encrypt the plain text message in the given scenario, use the following syntax –

$$C = P^e \text{ mod } n$$

Decryption Formula

The decryption process is very straightforward and includes analytics for calculation in a systematic approach. Considering receiver **C** has the private key **d**, the result modulus will be calculated as –

$$\text{Plaintext} = C^d \bmod n$$

Example

1. $P=7, Q=17$
2. $119=7*17$
3. $(7-1)*(17-1)=6*16=96$ factor 2 & 3, so $E=5$
4. $(D*5) \bmod (7-1)*(17-1)=1$, so $D=77$
5. $CT=10^5 \bmod 119 = 100000 \bmod 119 = 40$
6. Send 40
7. $PT=40^{77} \bmod 119 = 10$

Conclusion:

Thus we have studied how to implement RSA Algorithm

Oral Questions:

1. How is RSA algorithms implemented?
2. How to implement RSA key?
3. What are the vulnerabilities of RSA algorithm?
4. Is RSA a good or bad encryption system?
5. Why is RSA secure?

Lab Assignment No.	05
Title	Implementation of ECC algorithm
Roll No.	
Class	TE
Date of Completion	
Subject	Mini Project(Cyber Security)
Assessment Marks	
Assessor's Sign	

ASSIGNMENT No: 05

Title: Implementation of ECC algorithm

Problem Statement: Implementation of ECC algorithm

Prerequisite:

Basic of Python, Information security Algorithm

Software Requirements:

Python 3.7

Hardware Requirement:

2GB RAM, 500 GB HDD

Learning Objectives:

Learn ECC Algorithm

Outcomes:

After completion of this assignment students are able to understand the How ECC algorithm is used to encrypt and decrypt messages.

Theory:

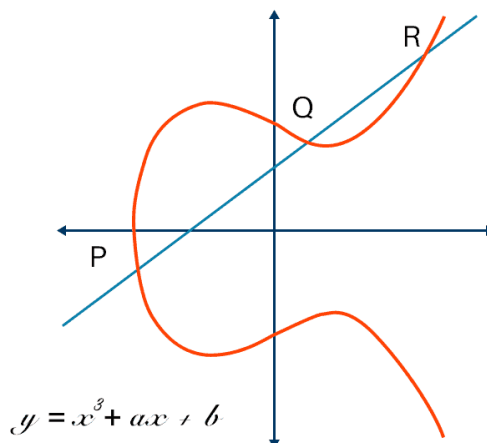
ECC, as the name implies, is an asymmetric encryption algorithm that employs the algebraic architecture of elliptic curves with finite fields.

Elliptic Curve Cryptography (ECC) is an encryption technology comparable to RSA that enables public-key encryption.

While RSA's security is dependent on huge prime numbers, ECC leverages the mathematical theory of elliptic curves to achieve the same level of security with considerably smaller keys.

Victor Miller and Neal Koblitz separately proposed elliptic curve ciphers in the mid-1980s.

On a high level, they are analogs of actual public cryptosystems in which modular arithmetic is substituted by elliptic curve operations.



Components of Elliptic Curve Cryptography

Below are the components of elliptic curve cryptography:

1. ECC keys:

- **Private key:** ECC cryptography's private key creation is as simple as safely producing a random integer in a specific range, making it highly quick. Any integer in the field represents a valid ECC private key.
- **Public keys:** Public keys within ECC are EC points, which are pairs of integer coordinates x , and y that lie on a curve. Because of its unique features, EC points can be compressed to a single coordinate + 1 bit (odd or even). As a result, the compressed public key corresponds to a 256-bit ECC.

2. Generator Point:

- ECC cryptosystems establish a special pre-defined EC point called generator point G (base point) for elliptic curves over finite fields, which can generate any other position in its subgroup over the elliptic curve by multiplying G from some integer in the range $[0 \dots r]$.
- The number r is referred to as the "ordering" of the cyclic subgroup.
- Elliptic curve subgroups typically contain numerous generator points, but cryptologists carefully select one of them to generate the entire group (or subgroup), and is excellent for performance optimizations in calculations. This is the " G " generator.

Elliptic Curve Cryptography Algorithms

Based on the arithmetic of elliptic curves over finite fields, Elliptic-Curve Cryptography (ECC) provides numerous sets of algorithms:

Digital signature algorithms:

- **Elliptic Curve Digital Signature Algorithm (ECDSA):** ECDSA, or Elliptic Curve Digital Signature Algorithm, is a more highly complicated public-key cryptography encryption algorithm. Elliptic curve cryptography is a type of public key cryptography that uses the algebraic structure of elliptic curves with finite fields as its foundation. Elliptic curve cryptography is primarily used to generate pseudo-random numbers, digital signatures, and other data.
- **Edwards-curve Digital Signature Algorithm (EdDSA):** The Edwards-curve Digital Signature Algorithm (EdDSA) was proposed as a replacement for the Elliptic Curve Digital

Signature Algorithm for performing fast public-key digital signatures (ECDSA). Its primary benefits for embedded devices are higher performance and simple, secure implementations. During a signature, no branch or lookup operations based on the secret values are performed. Many side-channel attacks are foiled by these properties.

Encryption algorithms:

- **Elliptic Curve Integrated Encryption Scheme (ECIES):** ECIES is a public-key authenticated encryption scheme that uses a KDF (key-derivation function) to generate a separate Medium Access Control key and symmetric encryption key from the ECDH shared secret. Because the ECIES algorithm incorporates a symmetric cipher, it can encrypt any amount of data. In practice, ECIES is used by standards such as Intelligent Transportation Systems.
- **EC-based ElGamal Elliptic Curve Cryptography:** ElGamal Elliptic Curve Cryptography is the public key cryptography equivalent of ElGamal encryption schemes that employ the Elliptic Curve Discrete Logarithm Problem. ElGamal is an asymmetric encryption algorithm that is used to send messages securely over long distances. Unfortunately, if the encrypted message is short enough, the algorithm is vulnerable to a Meet in the Middle attack.

Key Agreement algorithm:

- **Elliptic-curve Diffie-Hellman (ECDH):** Elliptic-curve Diffie-Hellman (ECDH) is a key agreement protocol that enables two parties to establish a shared secret over an insecure channel, each with an elliptic-curve public-private key pair. This shared secret can be used directly as a key or to generate another key. Following that, the key, or the derived key, can be used to encrypt subsequent communications with a symmetric-key cipher.
- **Fully Hashed Menezes-Qu-Vanstone(FHMQV):** Fully Hashed Menezes-Qu-Vanstone is an authenticated key agreement protocol based on the Diffie-Hellman scheme. MQV, like other authenticated Diffie-Hellman schemes, protects against an active attacker. The protocol can be adapted to work in any finite group, most notably elliptic curve groups, in which it is recognized as elliptic curve MQV (ECMQV).

Application of Elliptic Curve Cryptography

- **Diffie-Hellman:** The basic public-key cryptosystem suggested for secret key sharing is the Diffie-Hellman protocol. If A (Alice) and B (Bob) initially agree on a given curve, field size, and mathematical type. They then distribute the secret key in the following manner. We can see that all we need to build the Diffie-Hellman protocol is scalar multiplication.
- **Elliptic Curve Digital Signature Algorithm (ECDSA):** ECC is one of the most widely utilized digital signature implementation approaches in cryptocurrencies. In order to sign transactions, both Bitcoin and Ethereum use the field inverse multiplication, but also arithmetic multiplication, inverse function, and modular operation.
- **Online application:** Moreover, ECC is not limited to cryptocurrencies. It is an encryption standard that will be utilized by most online apps in the future due to its reduced key size and efficiency. Most commonly used in cryptocurrencies such as Bitcoin and Ethereum, along with single-way encryption of emails, data, and software.
- **Blockchain application:** The cryptocurrency Bitcoin employs elliptic curve cryptography. Ethereum 2.0 makes heavy use of elliptic curve pairs with BLS signatures, as stated in the IETF proposed BLS specification, to cryptographically ensure that a specific Eth2 validator has really verified a specific transaction.

Conclusion:

Thus we have studied how to implement ECC Algorithm

Oral Questions:

1. How do you implement ECC?
2. What are the applications of ECC?

Lab Assignment No.	06
Title	Enable/Configure (windows/ubuntu) firewall. Create rules to filter network traffic and to block unauthorized network traffic.
Roll No.	
Class	TE
Date of Completion	
Subject	Mini Project(Cyber Security)
Assessment Marks	
Assessor's Sign	

ASSIGNMENT No: 06

Title: Enable/Configure (windows/ubuntu) firewall. Create rules to filter network traffic and to block unauthorized network traffic.

Problem Statement: Enable/Configure (windows/ubuntu) firewall. Create rules to filter network traffic and to block unauthorized network traffic.

Prerequisite:

Basic of Python, Information security Algorithm

Software Requirements:

Python 3.7

Hardware Requirement:

2GB RAM, 500 GB HDD

Learning Objectives:

Learn ECC Algorithm

Outcomes:

After completion of this assignment students are able to understand the How to create rules to filter network traffic and to block unauthorized network traffic.

Theory:

Configuring firewalls in both Windows and Ubuntu involves setting rules to control incoming and outgoing network traffic. Here's an overview of how to enable/configure firewalls and create rules to filter and block unauthorized network traffic on both systems:

Windows Firewall:

Enabling Windows Firewall:

1. Open Windows Defender Security Center.
2. Go to Firewall & network protection.
3. Click on Turn Windows Firewall on or off.
4. Select Turn on Windows Firewall for both private and public networks.

Creating Firewall Rules:

1. Open Windows Defender Security Center.

2. Go to Firewall & network protection.
3. Click on Advanced settings.
4. In the Windows Firewall with Advanced Security window, you'll find inbound and outbound rules.
5. To create a new rule, right-click on either Inbound Rules or Outbound Rules, and select New Rule.
6. Follow the wizard to define the rule based on criteria like program, port, IP address, etc.
7. For blocking unauthorized traffic, create rules to block specific ports or programs that you don't want to allow through the firewall.

Ubuntu Firewall (UFW - Uncomplicated Firewall):

Enabling UFW:

1. Open a terminal.
2. Check UFW status with `sudo ufw status`.
3. If it's inactive, enable it with `sudo ufw enable`.

Creating Firewall Rules:

1. Open a terminal.
2. Use `sudo ufw [allow/deny] [port_number]` to allow or deny traffic on a specific port.
3. Use `sudo ufw allow from [IP_address] to any port [port_number]` to allow traffic from a specific IP address to a port.
4. For more complex rules, edit the UFW rules file directly at `/etc/ufw/rules`.

Conclusion:

Thus we have studied how to Enable/Configure (windows/ubuntu) firewall.

Oral Questions:

1. How do I enable firewall in Ubuntu Server?
2. How would you configure a firewall on a Linux server to ensure that only authorized traffic is allowed through?
3. How do I block a website in Ubuntu firewall?
4. How do I know if my firewall is enabled Ubuntu?

Lab Assignment No.	07
Title	Configure and demonstrate an Intrusion Detection System (IDS) to detect suspicious activities and generate alerts when detected.
Roll No.	
Class	TE
Date of Completion	
Subject	Mini Project(Cyber Security)
Assessment Marks	
Assessor's Sign	

ASSIGNMENT No: 07

Title: Configure and demonstrate an Intrusion Detection System (IDS) to detect suspicious activities and generate alerts when detected.

Problem Statement: Configure and demonstrate an Intrusion Detection System (IDS) to detect suspicious activities and generate alerts when detected.

Prerequisite:

Basic of Python, Information security Algorithm

Software Requirements:

Python 3.7

Hardware Requirement:

2GB RAM, 500 GB HDD

Learning Objectives:

Learn ECC Algorithm

Outcomes:

After completion of this assignment students are able to understand the How detection of suspicious activity is done and how alerts are generated.

Theory:

Types of IDS:

1. Host-based IDS (HIDS): Monitors activities on individual hosts or devices, analyzing log files and system calls for signs of malicious activity.
2. Network-based IDS (NIDS): Monitors network traffic in real-time, analyzing packets and looking for patterns that indicate an attack.

Detection Methods:

1. Signature-based Detection: Matches patterns in network traffic or system logs against known signatures of known threats.
2. Anomaly-based Detection: Establishes a baseline of normal activity and flags deviations from this baseline as potential intrusions.
3. Behavioral-based Detection: Analyzes the behavior of users and systems to identify abnormal or suspicious activities.

IDS Components:

1. Sensors/Agents: Capture and analyze network traffic or system activities.

2. **Analyzers:** Process data collected by sensors/agents and apply detection algorithms to identify potential threats.
3. **Alerting Mechanism:** Generates alerts when suspicious activities are detected, notifying system administrators or security personnel.
4. **Central Management Console:** Provides a centralized interface for monitoring and managing the IDS system.

Configuration and Demonstration:

For demonstration purposes, let's set up an open-source Network Intrusion Detection System (NIDS) using Suricata on an Ubuntu system:

1. **Install Suricata:**

Copy code

```
sudo apt-get install suricata
```

2. **Configure Suricata:**

- Update Suricata configuration file (/etc/suricata/suricata.yaml) to define rulesets, logging options, and network interfaces to monitor.

3. **Start Suricata:**

Copy code

```
sudo suricata -c /etc/suricata/suricata.yaml -i <interface>
```

4. **Generate Traffic:**

- Simulate network traffic, which can include legitimate traffic and malicious activities. Tools like Nmap or Metasploit can be used for this purpose.

5. **Monitor Alerts:**

- Suricata will generate alerts based on the configured rulesets and detection methods.
- View alerts in Suricata's log files (/var/log/suricata/fast.log, /var/log/suricata/eve.json) or using Suricata's provided tools.

6. **Respond to Alerts:**

- Investigate alerts to determine the nature of the detected activity.
- Take appropriate action, such as blocking suspicious IP addresses, updating firewall rules, or notifying relevant personnel.

Elliptic curve cryptography (ECC) is a public key cryptographic algorithm used to perform critical security functions, including encryption, authentication, and digital signatures.

Conclusion: Configuring and demonstrating an IDS involves setting up the IDS system,

configuring it to monitor network or system activities, generating simulated traffic to trigger alerts, and responding to alerts accordingly

Mini Project(Cloud Computing)

Laboratory

Course Code	Course Name	Teaching Scheme (Hrs./ Week)	Credits
317536	Mini Project(Cloud computing)	2	1

Course Objectives:

- To Learn AWS environment.
- To Learn Amazon RDS.
- To design and develop different applications using Amazon Services.

Course Outcomes:

On completion of the course, learner will be able to–

- CO1: Understanding of AWS environment.
- CO2:Understand Amazon RDS
- CO3:Understand and use of AWS Light sail

Table of Contents

Sr. No	Title of Experiment(Cloud Computing)	CO Mapping	Page No
1	Setting up AWS Environment: Create a new AWS account, Secure the root user, Create an IAM user to use in the account Set up the AWS CLI, Set up a Cloud9 environment.	CO1	
2	Setup, Create and visualize data in an Amazon Relational Database (Amazon RDS) MS SQL Express server using Amazon Quick Sight.	CO1, CO2	
3	Setup, Create and connect your Word Press site to an object storage bucket using Light sail service	CO3	

Lab Assignment No.	01
Title	Setting up AWS Environment: Create a new AWS account, Secure the root user, Create an IAM user to use in the account Set up the AWS CLI, Set up a Cloud9 environment
Roll No.	
Class	TE
Date of Completion	
Subject	Mini Project(Cloud Computing)
Assessment Marks	
Assessor's Sign	

ASSIGNMENT No: 01

Title: Setting up AWS Environment: Create a new AWS account, Secure the root user, Create an IAM user to use in the account Set up the AWS CLI, Set up a Cloud9 environment

Problem Statement: creating a new AWS account.

Prerequisite:

Database Management System

Software Requirements:

Internet browser

Hardware Requirements:

PIV, 2GB RAM, 500 GB HDD

Learning Objectives:

Learn setting up AWS Environment

Outcomes:

After completion of this assignment students are able to understand Setting up AWS Environment

Theory:

1. To create a new AWS account, go to aws.amazon.com and choose - Create an AWS Account.
2. First, choose an email address and an account name. Choose Verify email address. You will get a verification code in your email. Enter the verification code and choose Verify.
3. Create your root user password. Once you have entered and confirmed your password, choose Continue.

Sign up for AWS

Root user email address

Used for account recovery and some administrative functions

AWS account name

Choose a name for your account. You can change this name in your account settings after you sign up.

Verify email address

OR

Sign in to an existing AWS account

Sign up for AWS

Create your password

✔ It's you! Your email address has been successfully verified. ✕

Your password provides you with sign in access to AWS, so it's important we get it right.

Root user password

Confirm root user password

Continue (step 1 of 5)

OR

Sign in to an existing AWS account

4. Add contact information.
5. Add payment method.
6. Confirm Identity.

Sign up for AWS

Billing Information

Credit or Debit card number



AWS accepts all major credit and debit cards. To learn more about payment options, review our [FAQ](#)

Expiration date

Cardholder's name

Billing address

☒ Use my contact address

☐ Use a new address

Verify and Continue (step 3 of 5)

You might be redirected to your bank's website to authorize the verification charge.

Sign up for AWS

Confirm your identity

Before you can use your AWS account, you must verify your phone number. When you continue, the AWS automated system will contact you with a verification code.

How should we send you the verification code?

- ☒ Text message (SMS)
☐ Voice call

Country or region code

Mobile phone number

Security check

	🔊
	🔄

Type the characters as shown above

Send SMS (step 4 of 5)

1. Select a support plan and complete Signup.

Sign up for AWS

Select a support plan

Choose a support plan for your business or personal account. [Compare plans and pricing examples](#). You can change your plan anytime in the AWS Management Console.

☒ Basic support - Free • Recommended for new users just getting started with AWS • 24x7 self-service access to AWS resources • For account and billing issues only • Access to Personal Health Dashboard & Trusted Advisor 	☐ Developer support - From \$29/month • Recommended for developers experimenting with AWS • Email access to AWS Support during business hours • 12 (business)-hour response times 	☐ Business support - From \$100/month • Recommended for running production workloads on AWS • 24x7 tech support via email, phone, and chat • 1-hour response times • Full set of Trusted Advisor best-practice recommendations 
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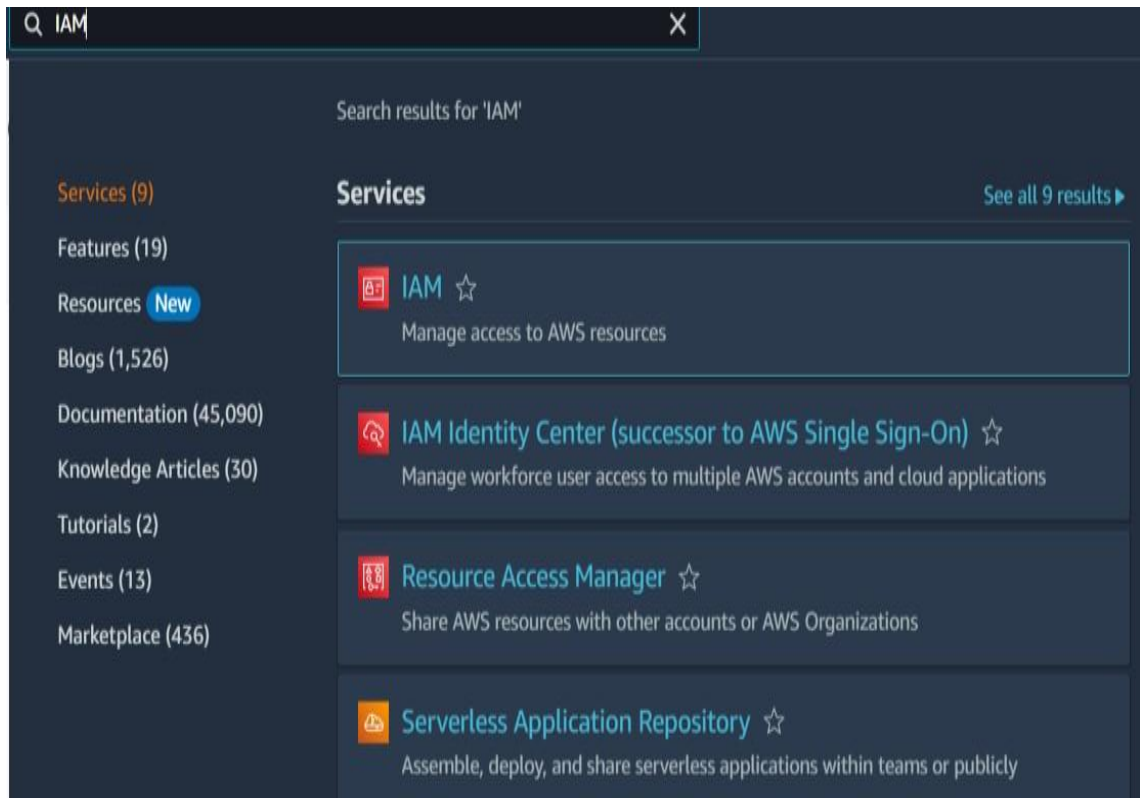


Need Enterprise level support?

From \$15,000 a month you will receive 15-minute response times and concierge-style experience with an assigned Technical Account Manager. [Learn more](#)

[Complete sign up](#)

2. In AWS Management Console, click on the search bar and enter IAM, and then select IAM. You will be redirected to the IAM dashboard.



3. Under security recommendations, choose **Add MFA**.

Identity and Access Management (IAM)

Dashboard

▼ Access management

- User groups
- Users
- Roles
- Policies
- Identity providers
- Account settings

▼ Access reports

- Access analyzer
- Archive rules
- Analyzers

IAM dashboard

Security recommendations 2

⚠️ Add MFA for root user

Add MFA for root user - Enable multi-factor authentication (MFA) for the root user to improve security for this account.

Add MFA

✅ Root user has no active access keys

Using access keys attached to an IAM user instead of the root user improves security.

⚠️ Update your access permissions for AWS Billing, Cost Management, and Account consoles

We are replacing the following IAM actions for Billing, Cost Management, and Account consoles with granular IAM actions: aws-portal:ViewBilling, aws-portal:ModifyBilling, aws-portal:ViewAccount, aws-portal:ModifyAccount, aws-portal:ViewPaymentMethods, aws-portal:ModifyPaymentMethods, aws-portal:ViewUsage, purchase-orders:ViewPurchaseOrders, and purchase-orders:ModifyPurchaseOrders. To ensure you don't lose access to AWS Billing, Cost Management, and Account console based features, update your existing IAM policies to include the new IAM actions before July 2023. Examples of features impacted include AWS Cost Explorer, AWS Budgets, Billing console, and more. For more information, please visit [blog](#).

View affected policies

AWS Account

Account ID

553522802144

Account Alias

553522802144 Create

Sign-in URL for IAM users in this account

<https://553522802144.signin.aws.amazon.com/console>

Quick Links

[My security credentials](#)

Manage your access keys, multi-factor authentication (MFA) and other credentials.

Tools

IAM resources

4. Select a virtual MFA device.

5. When everything is ready, choose Assign MFA. Your root user is now secure.

6. Install AWS CLI version 2

7. Setup AWS CLI

8. Open IAM dashboard and select **users**.

IAM resources

User groups
0

Users
1

Roles
4

Policies
0

Identity providers
0

9. Select **Add User**.

IAM > Users

Users (1) Info

An IAM user is an identity with long-term credentials that is used to interact with AWS in an account.

Find users by username or access key

User name

Groups

Last activity

MFA

Password age

Active key age

nrg2307

None

Never

None

None

-

10. Add new user

Specify user details

User details

User name

user1

The user name can have up to 64 characters. Valid characters: A-Z, a-z, 0-9, and + = , _ @ - (hyphen)

☐

Provide user access to the AWS Management Console - optional

If you're providing console access to a person, it's a best practice to manage their access in IAM Identity Center.

If you are creating programmatic access through access keys or service-specific credentials for AWS CodeCommit or Amazon Keyspaces, you can generate them after you create this IAM user. [Learn more](#)

IAM > User groups > Create user group

Create user group

Name the group

User group name

Enter a meaningful name to identify this group.

administrators

Maximum 128 characters. Use alphanumeric and '+ = , _ @ -' characters.

Add users to the group - Optional (0) Info

An IAM user is an entity that you create in AWS to represent the person or application that uses it to interact with AWS. A user can belong to up to 10 groups.

Search

User name

Groups

Last activity

Creation time

No resources to display

Attach permissions policies - Optional (Selected 1/767) Info

You can attach up to 10 policies to this user group. All the users in this group will have permissions that are defined in the selected policies.

Filter policies by property or policy name and press enter

4 matches

administratoraccess

Clear filters

Policy name

Type

Description

AdministratorAccess

AWS managed - job functi...

Provides full access

AdministratorAccess-Amplify

AWS managed

Grants account adn

AdministratorAccess-AWSElasticBeanstalk

AWS managed

Grants account adn

AWSAuditManagerAdministratorAccess

AWS managed

Provides administra

Cancel

Create group

[Alt+S]

Global

Nrj

Permissions options

☒ Add user to group

Add user to an existing group, or create a new group. We recommend using groups to manage user permissions by job function.

☐ Copy permissions

Copy all group memberships, attached managed policies, and inline policies from an existing user.

☐ Attach policies directly

Attach a managed policy directly to a user. As a best practice, we recommend attaching policies to a group instead. Then, add the user to the appropriate group.

User groups (1/2)

Search groups

< 1 >

	Group name	Users	Attached policies	Created
☐	administrators	0	AdministratorAccess	2023-03-26 (16 minutes ago)
☒	administrators1	0	AdministratorAccess	2023-03-26 (8 minutes ago)

Permissions boundary - optional

Set a permissions boundary to control the maximum permissions for this user. Use this advanced feature used to delegate permission management to others. [Learn more](#)

Cancel

Previous

Next

Review and create

Review your choices. After you create the user, you can view and download the autogenerated password, if enabled.

User details

User name

user1

Console password type

None

Require password reset

No

Permissions summary

Name	Type	Used as
administrators1	Group	Permissions group

Tags - optional

Tags are key-value pairs you can add to AWS resources to help identify, organize, or search for resources. Choose any tags you want to associate with this user.

Key

admin

X

Value - optional

Enter value

Remove

Add new tag

You can add up to 49 more tags.

Cancel

Previous

Create user

[Alt+S]

Global

Nrj

IAM > Users

Users (2) Info

Delete

Add users

Find users by username or access key

< 1 >

	User name	Groups	Last activity	MFA	Password age	Active key age
☐	nj2307	None	Never	None	None	-
☐	user1	administrators1	Never	None	None	-

Click on create access key

48

Access key best practices & alternatives

Avoid using long-term credentials like access keys to improve your security. Consider the following use cases and alternatives.

☒ **Command Line Interface (CLI)**
You plan to use this access key to enable the AWS CLI to access your AWS account.

☐ **Local code**
You plan to use this access key to enable application code in a local development environment to access your AWS account.

☐ **Application running on an AWS compute service**
You plan to use this access key to enable application code running on an AWS compute service like Amazon EC2, Amazon ECS, or AWS Lambda to access your AWS account.

☐ **Third-party service**

Access key created
This is the only time that the secret access key can be viewed or downloaded. You cannot recover it later. However, you can create a new access key any time.

IAM > Users > user1 > Create access key

Step 1
Access key best practices & alternatives

Step 2 - optional
Set description tag

Step 3
Retrieve access keys

Retrieve access keys

Access key
If you lose or forget your secret access key, you cannot retrieve it. Instead, create a new access key and make the old key inactive.

Access key	Secret access key
AKIAYBYEHVXQJALSPT5G	***** Show

Access key best practices

- Never store your access key in plain text, in a code repository, or in code.
- Disable or delete access key when no longer needed.
- Enable least-privilege permissions.
- Rotate access keys regularly.

For more details about managing access keys, see the [Best practices for managing AWS access keys](#).

Access keys (1)

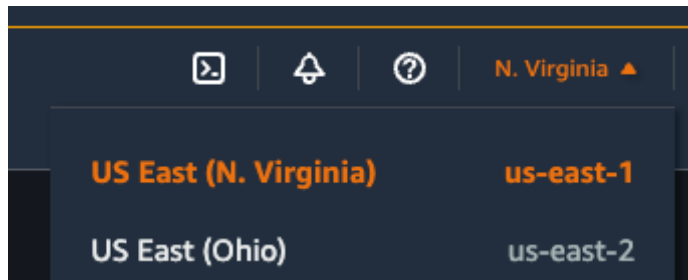
Use access keys to send programmatic calls to AWS from the AWS CLI, AWS Tools for PowerShell, AWS SDKs, or direct AWS API calls. You can have a maximum of 2 (inactive) at a time. [Learn more](#)

Create access key

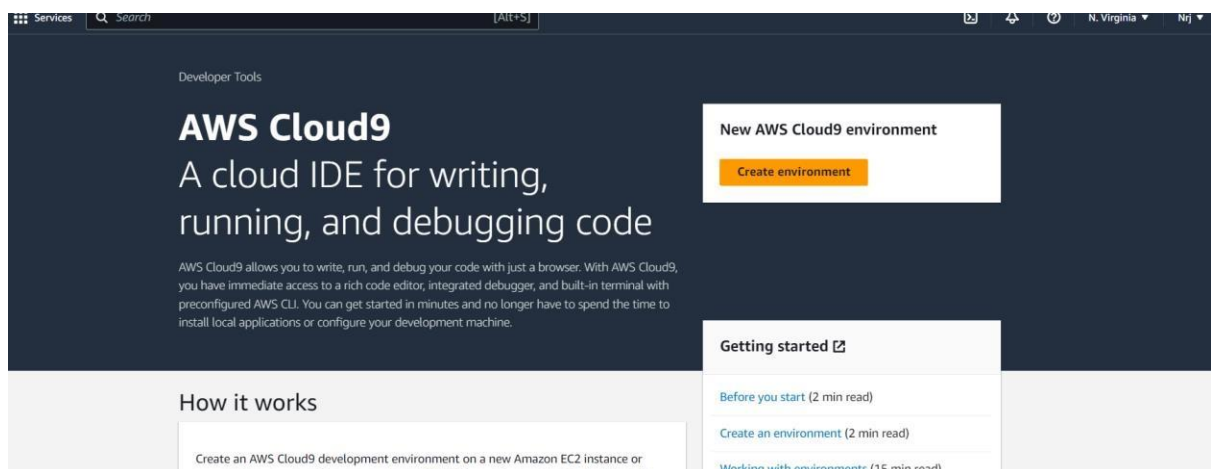
AKIAYBYEHVXQA4S6PH57	
Description	Status
key1	Active
Last used	Created
None	3 minutes ago
Last used region	Last used service
N/A	N/A

11. Use the AWS CLI to create an AWS Cloud9 development environment.
12. To open your environment, go to console.aws.amazon.com/cloud9.

13. In the top navigation bar, choose the AWS Region where your environment is located. Make sure you are logged in with the same user you have configured in your AWS CLI.

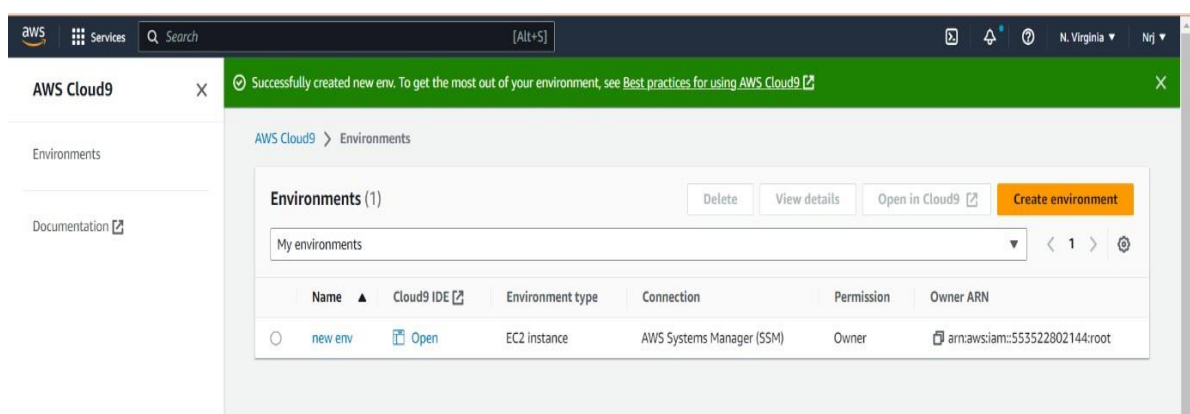


14. Create a new environment



15. The new environment gets successfully created.

16. In the list of environments, find the one you want to open and choose Open IDE.



Oral Questions:

1. What are the basics of AWS?
2. What is AWS in cloud computing?
3. What are main components of AWS?

Lab Assignment No.	02
Title	Setup, Create and visualize data in an Amazon Relational Database (Amazon RDS) MS SQL Express server using Amazon Quick Sight.
Roll No.	
Class	TE
Date of Completion	
Subject	Mini Project(Cloud Computing)
Assessment Marks	
Assessor's Sign	

ASSIGNMENT No: 02

Title: Setup, Create and visualize data in an Amazon Relational Database (Amazon RDS) MS SQL Express server using Amazon Quick Sight.

Problem Statement: Setup, Create and visualize data in an Amazon Relational Database (Amazon RDS) MS SQL Express server using Amazon Quick Sight.

Prerequisite:

Database Management System

Software Requirements:

Internet browser

Hardware Requirements:

PIV, 2GB RAM, 500 GB HDD

Learning Objectives:

Learn setting up AWS Environment

Outcomes:

After completion of this assignment students are able to understand to Setup, Create and visualize data in an Amazon Relational Database (Amazon RDS) MS SQL Express server using Amazon Quick Sight.

Theory:

Step 1. Sign in to an AWS Account

Step 2. Create a Microsoft SQL Server Express Edition database in Amazon RDS

The screenshot shows the Amazon RDS console interface. At the top, the AWS logo and 'Services' dropdown are visible. A search bar contains the text 'Search for services, features, marketplace products, and docs'. The top right shows the user 'Administrator_Access/' and the region 'N. Virginia'. The left sidebar lists navigation options: Dashboard, Databases, Query Editor, Performance Insights, Snapshots, Automated backups, Reserved instances, Proxies, Subnet groups, Parameter groups, Option groups, Custom Availability Zones, Events, Event subscriptions, Recommendations, and Certificate update. The main content area is titled 'Amazon RDS' and includes a 'Resources' section with a 'Refresh' button. Below this, it states 'You are using the following Amazon RDS resources in the US East (N. Virginia) region (used/quota)'. A table lists various resources and their counts: DB Instances (0/40), Parameter groups (0), DB Clusters (0/40), Option groups (0), Reserved Instances (0/40), Subnet groups (0/50), Snapshots (0), Supported platforms VPC, Manual (0/100), and Default network vpc-14f6f072. Below the table is the 'Create database' section, which includes a description of Amazon RDS and two buttons: 'Restore from S3' and 'Create database'. The 'Create database' button is highlighted with a red box. At the bottom of the 'Create database' section, a note states: 'Note: your DB instances will launch in the US East (N. Virginia) region'. On the right side of the console, there are several 'Recommended for you' cards, including 'Migrate SSRS to RDS for SQL Server', 'Build RDS Operational Tasks', 'Backup and Restore Using AWS Backup', and 'Time-Series Tables in PostgreSQL'. An 'Additional information' section is also visible at the bottom right.

On the Create database page, in the Choose a database creation method section, choose Easy Create.

In the Configuration section, make the following changes:

- For Engine type, choose Microsoft SQL Server.
- For DB instance size, choose Free tier.
- For DB instance identifier, type qsdatabase.
- For Master username, enter admin.
- For Master password, type a unique password, and confirm password.

RDS > Create database

Create database

Choose a database creation method [Info](#)

☐ Standard create

You set all of the configuration options, including ones for availability, security, backups, and maintenance.

☒ Easy create

Use recommended best-practice configurations. Some configuration options can be changed after the database is created.

Configuration

Engine type [Info](#)

☐ Amazon Aurora



☐ MySQL



☐ MariaDB



☐ PostgreSQL



☒ Microsoft SQL Server



DB instance size

☐ Production

db.r5.xlarge
4 vCPUs
32 GiB RAM
500 GiB
3.198 USD/hour

☐ Dev/Test

db.m5.large
2 vCPUs
8 GiB RAM
100 GiB
0.993 USD/hour

☒ Free tier

db.t2.micro
1 vCPUs
1 GiB RAM
20 GiB
0.025 USD/hour

DB instance identifier

Type a name for your DB instance. The name must be unique across all DB instances owned by your AWS account in the current AWS Region.

The DB instance identifier is case-insensitive, but is stored as all lowercase (as in "mydbinstance"). Constraints: 1 to 60 alphanumeric characters or hyphens. First character must be a letter. Can't contain two consecutive hyphens. Can't end with a hyphen.

Master username [Info](#)

Type a login ID for the master user of your DB instance.

1 to 16 alphanumeric characters. First character must be a letter

☐ Auto generate a password

Amazon RDS can generate a password for you, or you can specify your own password

Master password [Info](#)

Constraints: At least 8 printable ASCII characters. Can't contain any of the following: / (slash), '(single quote), "(double quote) and @ (at sign).

Confirm password [Info](#)

▼ View default settings for Easy create

Easy create sets the following configurations to their default values, some of which can be changed later. If you want to change any of these settings now, use [Standard Create](#).

Configuration ▼	Value	Editable after database is created ▲
Encryption	Enabled	No
VPC	Default VPC vpc-1a5b0c12	No
Option Group	default- sqlserver-ex-14-00	Yes
Subnet Group	default-vpc- 1a5b0c12	Yes
Automatic Backups	Enabled	Yes
VPC Security Group	sg-1a5b0c12	Yes
Publically Accessible	No	Yes
Database Port	1433	Yes
DB Instance Identifier	qsdatabase	Yes
DB Engine Version	14.00.3356.20.v1	Yes
DB Parameter Group	default.sqlserver-ex-14.0	Yes
Performance Insights	Enabled	Yes
Monitoring	Enabled	Yes
Maintenance	Auto Minor Version Upgrade Enabled	Yes
Delete Protection	Not Enabled	Yes

 You are responsible for ensuring that you have all of the necessary rights for any third-party products or services that you use with AWS services.

Cancel

Create database

Step 3. Download and connect to a Microsoft SQL Server client

Download SQL Server Management Studio (SSMS)

12/17/2020 • 6 minutes to read •  +35

Applies to:  SQL Server (all supported versions)  Azure SQL Database  Azure SQL Managed Instance  Azure Synapse Analytics

SQL Server Management Studio (SSMS) is an integrated environment for managing any SQL infrastructure, from SQL Server to Azure SQL Database. SSMS provides tools to configure, monitor, and administer instances of SQL Server and databases. Use SSMS to deploy, monitor, and upgrade the data-tier components used by your applications, and build queries and scripts.

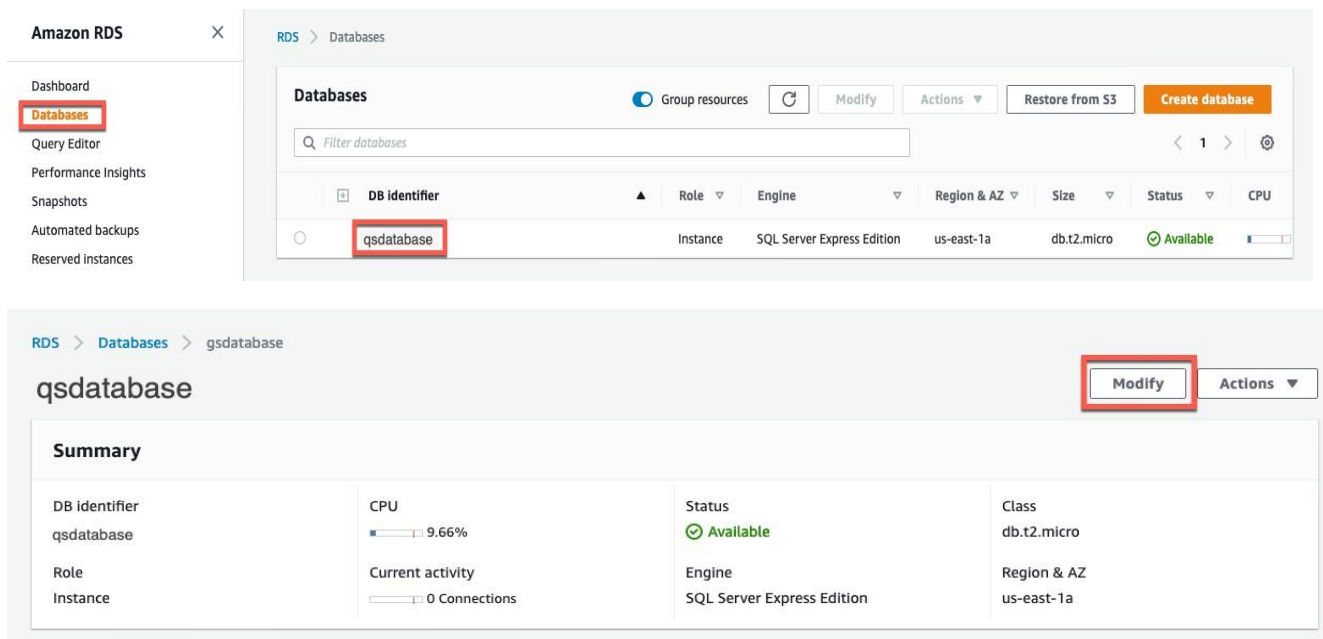
Use SSMS to query, design, and manage your databases and data warehouses, wherever they are - on your local computer, or in the cloud.

Download SSMS

 [Download SQL Server Management Studio \(SSMS\)](#) 

SSMS 18.8 is the latest general availability (GA) version of SSMS. If you have a previous GA version of SSMS 18 installed, installing SSMS 18.8 upgrades it to 18.8.

Open the Amazon RDS console



The screenshot shows the Amazon RDS console. On the left is a navigation menu with 'Databases' highlighted. The main area displays a list of databases. One database, 'gsdatabase', is highlighted with a red box. Below this, the 'gsdatabase' instance details page is shown. In the top right corner of this page, the 'Modify' button is highlighted with a red box.

Amazon RDS X

RDS > Databases

Databases      

Filter databases

	DB identifier	Role	Engine	Region & AZ	Size	Status	CPU
	gsdatabase	Instance	SQL Server Express Edition	us-east-1a	db.t2.micro	Available	

RDS > Databases > gsdatabase

gsdatabase  

Summary

DB identifier gsdatabase	CPU  9.66%	Status  Available	Class db.t2.micro
Role Instance	Current activity  0 Connections	Engine SQL Server Express Edition	Region & AZ us-east-1a

On the ModifyDB instance: gsdatabase page, in the Connectivity section, choose Additional Configuration. Then, choose Publicly accessible, and choose Continue.

Connectivity

Subnet group

default-vpc-14f6f072 ▼

Security group

List of DB security groups to associate with this DB instance.

Choose security groups ▼

default ✕

Certificate authority

rds-ca-2019 ▼

▼ Additional configuration

Public access

☒ Publicly accessible
EC2 instances and devices outside the VPC can connect to the instance. You define the security groups for supported devices and instances.

☐ Not publicly accessible
No IP address is assigned to the DB instance. EC2 instances and devices outside the VPC can't connect.

Database port

Specify the TCP/IP port that the DB instance will use for application connections. The application connection string must specify the port number. The DB security group and your firewall must allow connections to the port. [Learn more](#) 

1433 ▼

Cancel 

On the ModifyDB instance: qsdatabase page, in the Scheduling of modifications section, choose Apply immediately. Then, choose Modify DBinstance.

On the left-hand navigation, choose Databases. Then, choose qsdatabase.

On the qsdatabase page, in the Connectivity & security section, choose the VPC security groups link.

Connectivity & security

Endpoint & port

Endpoint
qsdatabase.c-xxxxxx.us-east-1.rds.amazonaws.com

Port
1433

Networking

Availability zone
us-east-1a

VPC
vpc-xxxxxx

Subnet group
default-vpc-xxxxxx

Subnets
subnet-xxxxxx
subnet-xxxxxx
subnet-xxxxxx
subnet-xxxxxx
subnet-xxxxxx

Security

VPC security groups
default (sg-xxxxxx) (active)

Public accessibility
Yes

Certificate authority
rds-ca-2019

Certificate authority date
August 22, 2024 13:08

On the Security groups page, choose the Security group ID.

On the sg-default page, in the Inbound rules section, choose Edit inboundrules.

Edit inbound rules [Info](#)

Inbound rules control the incoming traffic that's allowed to reach the instance.

Inbound rules [Info](#)

Type Info	Protocol Info	Port range Info	Source Info	Description - optional Info
All TCP	TCP	0 - 65535	My IP	

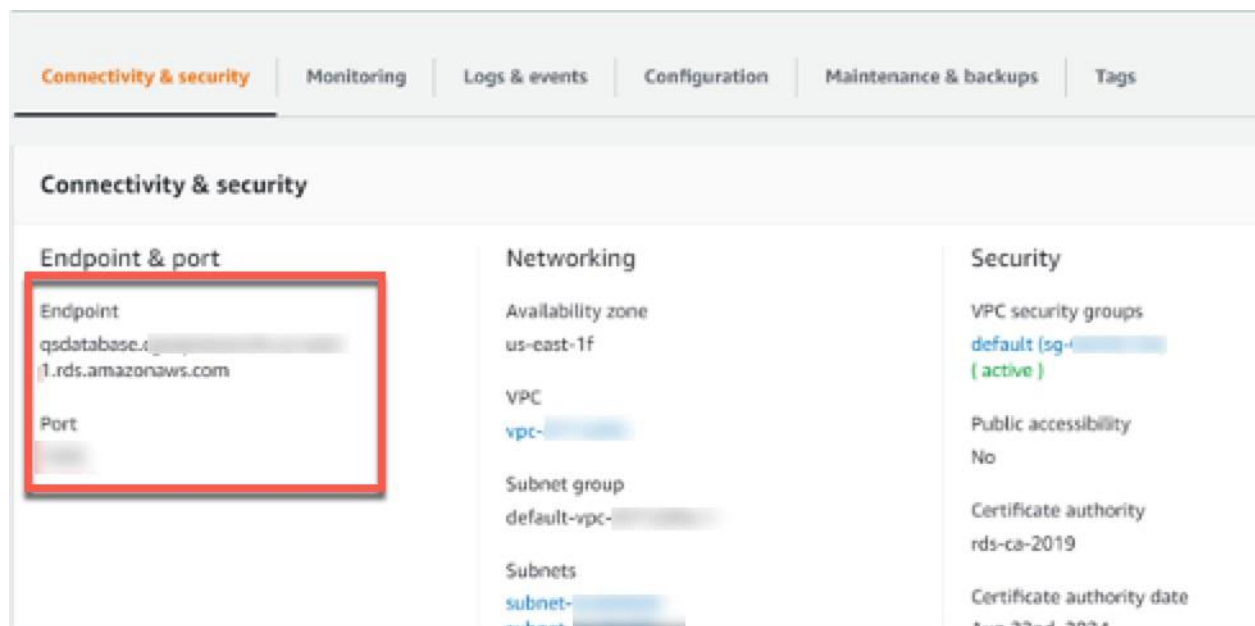
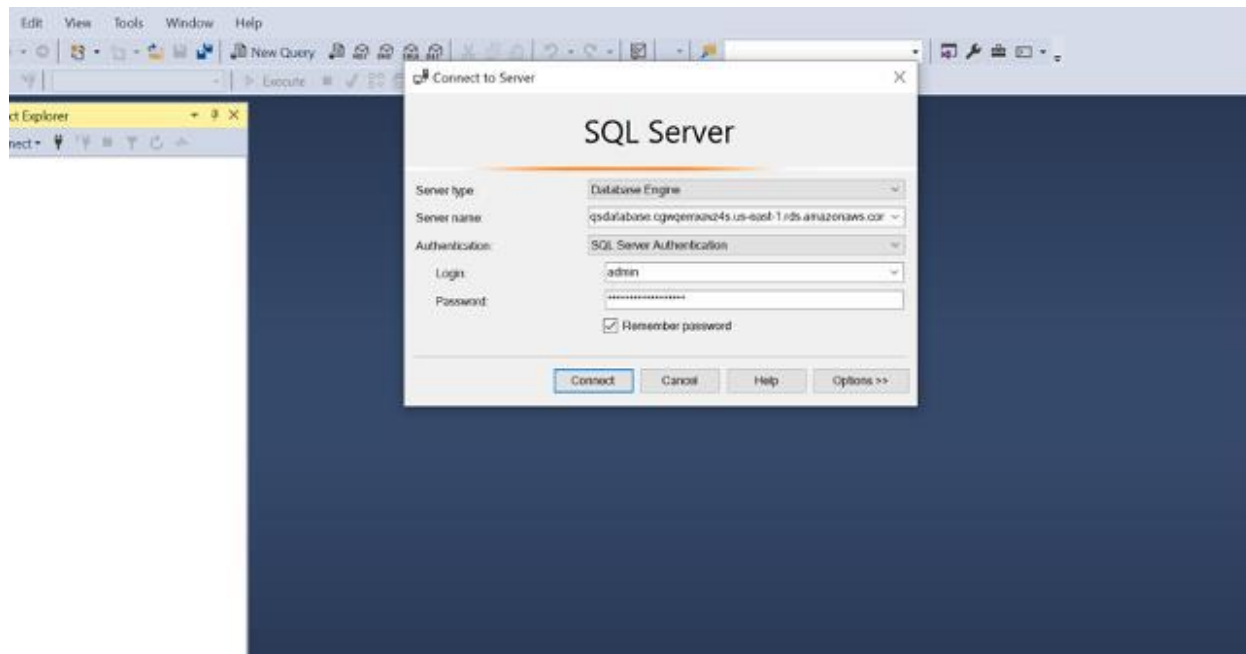
[Add rule](#)

NOTE: Any edits made on existing rules will result in the edited rule being deleted and a new rule created with the new details. This will cause traffic that depends on that rule to be dropped for a very brief period of time until the new rule can be created.

[Cancel](#) [Preview changes](#) [Save rules](#)

Verify that the SSMS Client download has completed. Then, install and open the software.

For Server Name, paste the qsdatabase Endpoint and Port separated by commas.



Step 4. Create a sample database and tables, and load sample data Open SQL Server Management Studio, in the left-hand navigation, choose Databases. Then, right click and choose Create Database.

On the New database page, for Database name, type Visualize. Then, choose OK.

Choose Visualize, and choose New Query.

In the Query editor, copy and paste the following script.

Once the script is successfully run, the tables will be created and loaded with the sample data.

```
CREATE TABLE newhire( empno
INT PRIMARY KEY,ename
VARCHAR(10),
job VARCHAR(9),
manager INT NULL,
hiredate DATETIME, salary
NUMERIC(7,2),
comm NUMERIC(7,2) NULL,
department INT)
begin
insert into newhire values
(1,'JOHNSON','ADMIN',6,'12-17-1990',18000,NULL,4)
insert into newhire values (2,'HARDING','MANAGER',9,'02-02-
1998',52000,300,3)
insert into newhire values
(3,'TAFT','SALES I',2,'01-02-1996',25000,500,3)
insert into newhire values
(4,'HOOVER','SALES I',2,'04-02-1990',27000,NULL,3)
insert into newhire values
(5,'LINCOLN','TECH',6,'06-23-1994',22500,1400,4)
insert into newhire values
(6,'GARFIELD','MANAGER',9,'05-01-1993',54000,NULL,4)
insert into newhire values
(7,'POLK','TECH',6,'09-22-1997',25000,NULL,4)
insert into newhire values
(8,'GRANT','ENGINEER',10,'03-30-1997',32000,NULL,2)
insert into newhire values
(9,'JACKSON','CEO',NULL,'01-01-1990',75000,NULL,4)
insert into newhire values
(10,'FILLMORE','MANAGER',9,'08-09-1994',56000,NULL,2)
insert into newhire values
(11,'ADAMS','ENGINEER',10,'03-15-1996',34000,NULL,2)
insert into newhire values
(12,'WASHINGTON','ADMIN',6,'04-16-1998',18000,NULL,4)
```

```

insert into newhire values
(13,'MONROE','ENGINEER',10,'12-03-2000',30000,NULL,2)
insert into newhire values
(14,'ROOSEVELT','CPA',9,'10-12-1995',35000,NULL,1)
end
CREATE TABLE department(deptno
INT NOT NULL, dname
VARCHAR(14),
loc VARCHAR(13))
begin
insert into department values (1,'ACCOUNTING','ST LOUIS') insert into
department values (2,'RESEARCH','NEW YORK') insert into department
values (3,'SALES','ATLANTA')
insert into department values (4, 'OPERATIONS','SEATTLE')end

```

Step 5. Make the database instance Not publicly accessible

- ☐ Open the Amazon RDS console, in the left-hand navigation, choose Databases. Then, choose the qsdatabase
- ☐ On the qsdatabase page, choose Modify.
- ☐ On the ModifyDB instance:qsdatabase page, in the Connectivity section, choose Additional Configuration. Then, choose Not publicly accessible, and choose Continue.
- ☐ On the ModifyDB instance:qsdatabase page, in the Scheduling of modifications section, choose Apply immediately. Then, choose Modify DB instance.

Step 6. Enable the RDS database instance for access to AmazonQuickSight

- ☐ Open the Amazon RDS console, in the left-hand navigation, choose Databases. Then, choose the qsdatabase.
- ☐ On the qsdatabase page, in the Connectivity & security section, copy the VPC id.
- ☐ Under Security, choose the VPC security groups link.
- ☐ On the Security Groups page, choose Create security group.
- ☐ On the Create security group page, in the Basic details section, enter the following details. For Name, type RDS SecGP For Description, type for QS For VPC, choose the VPC id for your RDS instance. Then, choose Create security group.

- ☐ On the Security Groups page, copy the Security group ID.
- ☐ On the Security Groups page, choose Create security group.

Create security group [Info](#)

A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.

Basic details

Security group name [Info](#)

Name cannot be edited after creation.

Description [Info](#)

VPC [Info](#)

Inbound rules [Info](#)

Type Info	Protocol Info	Port range Info	Source Info	Description - optional Info	
All traffic	All	All	Custom sg-ce- X		Delete

Outbound rules [Info](#)

Type Info	Protocol Info	Port range Info	Destination Info	Description - optional Info	
All traffic	All	All	Custom 0.0.0.0/0 X		Delete

Tags - optional
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

No tags associated with the resource.

You can add up to 50 more tag

On the sg-QS SecGp page, copy the security group id. This security group is needed for Amazon QuickSight to connect to Amazon RDS.

Security Groups (2) [Info](#)

	Name	Security group ID	Security group name	VPC ID	Description	Owner
☐	-	sg- sg-12345678901234567890123456789012	QS SecGp	vpc- vpc-12345678901234567890123456789012	for QS	arn:aws:iam::123456789012:role/QuickSightRole
☐	-	sg- sg-12345678901234567890123456789012	default	vpc- vpc-12345678901234567890123456789012	default VPC security gr...	arn:aws:iam::123456789012:role/DefaultVPCSecurityGroupRole

- ☐ In the Inbound rules section, choose Edit inbound rules.
- ☐ On the Edit inbound rules page, in the Inbound rules section, choose Add rule. Then, enter the following details.
For Type, choose MSSQL For
Source, choose Custom
In the search box, paste the security group id
- ☐ Choose Save rules. This security group is needed for Amazon RDS to connect Amazon QuickSight.
- ☐ Open the Amazon RDS console, in the left-hand navigation, choose Databases. Then, choose the qsdatabase.
- ☐ On the qsdatabase page, choose Modify.
- ☐ On the Modify DB instance: qsdatabase page, in the Connectivity section, for Security group, choose RDS SecGP (forQS). Then, choose Continue.
- ☐ On the Modify DB instance: qsdatabase page, in the Scheduling of modifications section, choose Apply immediately. Then, choose Modify DB instance.

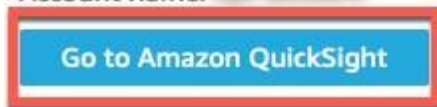
Step 7. Create your Amazon QuickSight account

- ☐ Open the Amazon QuickSight landing page, and choose Sign up for QuickSight.
- ☐ On the Create your QuickSight account page, for Edition, choose Enterprise, and choose Continue.
- ☐ On the Create your QuickSight account page, in the Edition section, choose Use IAM federated identities and QuickSight-managed users.
- ☐ In the QuickSight region section, enter the following details. Select a region from the drop-down list.
For QuickSight account name, type a unique account name.
For Notification email address, type an email address where you will receive notifications.
- ☐ Then, choose Finish.

Congratulations! You are signed up for Amazon QuickSight!

Access QuickSight with the following information

Account name: 



Step 8. Enable Amazon QuickSight to connect to Amazon RDS and create a dataset for visualization



- ☐ On the left navigation pane, choose Manage VPC connections. Then, choose Add VPC connection.
- ☐ In your web browser, open a new tab. Then, open the [Amazon RDS console](#), in the left-hand navigation, choose Databases. Then, choose the qsdatabase.
- ☐ On the qsdatabase page, in the Connectivity & security section, under VPC, copy the id. Then, under Subnets, copy one of the ids.

Adding VPC connection

You can connect QuickSight to your data secured by Virtual Private Cloud (VPCs). Enter the IDs for the VPC connection below. Use the links below to locate them in the AWS console. [Learn more](#)

- VPC ID [AWS VPC console](#)
- Subnet ID [AWS VPC subnet console](#)
- Security group [AWS security group console](#)
- DNS resolvers [Amazon Route 53 Resolver console](#)

VPC connection name

VPC ID

Subnet ID

Security group ID

DNS resolver endpoints (optional) ⓘ

VPC connection details cannot be changed later.

- ☐ On the top left corner of your screen, choose the QuickSight icon. Then, in the left navigation, choose Datasets.
- ☐ On the Datasets page, choose New dataset.
- ☐ On the Create a Datasets page, choose RDS.

New RDS data source ×

Data source name
DataFromRDS

Instance ID
qsdatabase ▼

Connection type
RDSVPC ▼

Database name
Visualize

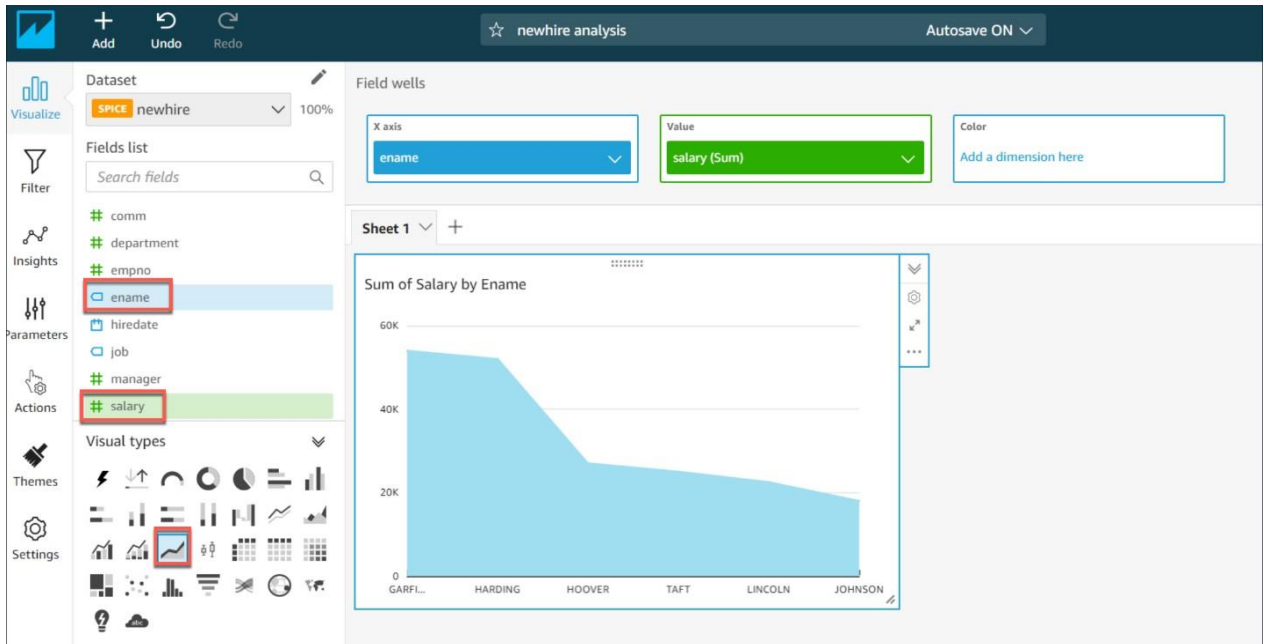
Username
admin

Password
.....

Validate connection SSL is enabled Create data source

- ☐ On the Choose your table page, in the Schema section, choose dbo.
- ☐ In the Tables section, choose newhire. Then, choose Select.
- ☐ On the Finish dataset creation page, leave the default selections, and choose Visualize.
- ☐ On the Visualize page, in the Visual types section, choose the Stacked Area Line Chart.

- In the Fields list section, drag and drop ename and salary to the Field Wells section.



Step 9. Clean up

- Delete the dashboard:
- Delete the analysis:
- Delete the data:
- Delete the database instance:
- Uninstall the MS SQL Management Studio client:

Oral Questions:

1. What is AWS relational database?
2. What is the use of RDS in AWS?
3. Which AWS service is a relational database compatible?
4. What is the structure of RDS?

Lab Assignment No.	03
Title	Setup, Create and connect your Word Press site to an object storage bucket using Lightsail service.
Roll No.	
Class	TE
Date of Completion	
Subject	Mini Project(Cloud Computing)
Assessment Marks	
Assessor's Sign	

ASSIGNMENT No: 03

Title: Setup, Create and connect your Word Press site to an object storage bucket using Lightsail service.

Problem Statement: . Setup, Create and connect your Word Press site to an object storage bucket using Lightsail service.

Prerequisite:

Database Management System

Software Requirements:

Internet browser

Hardware Requirements:

PIV, 2GB RAM, 500 GB HDD

Learning Objectives:

Learn setting up AWS Environment

Outcomes:

After completion of this assignment students are able to understand the Setup, Create and connect your Word Press site to an object storage bucket using Lightsail service.

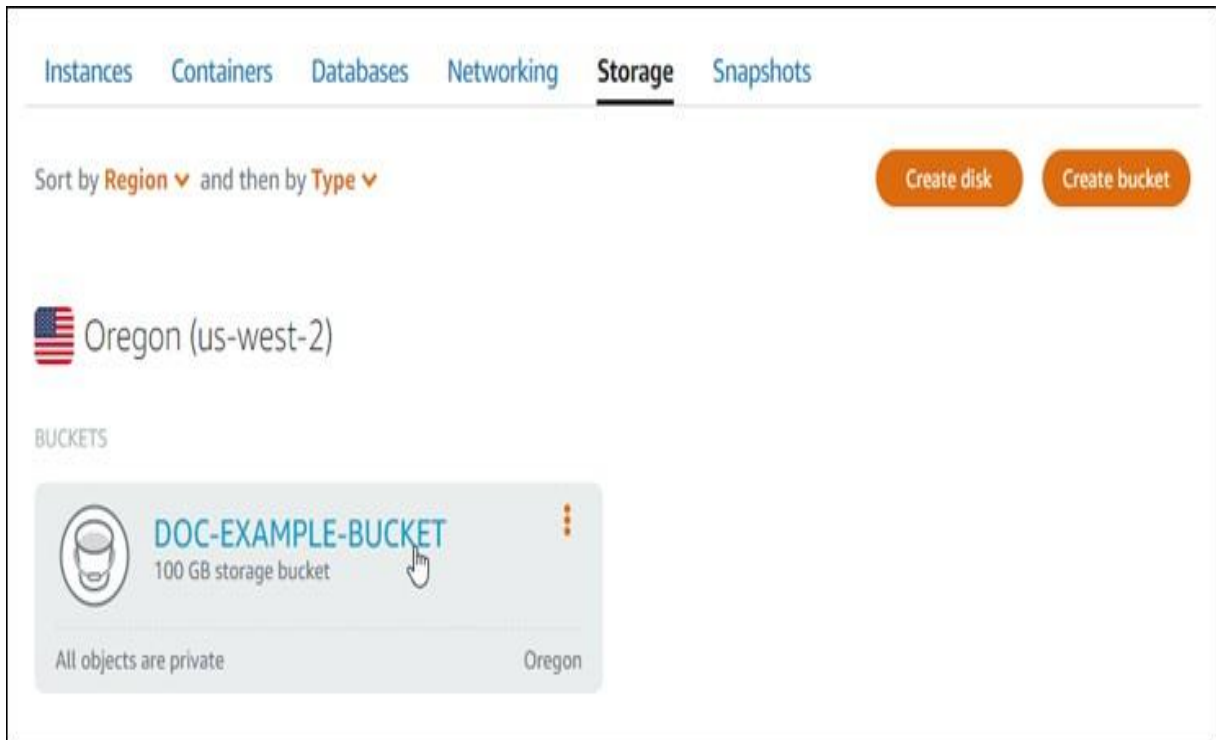
Theory:

Step 1: Prerequisites

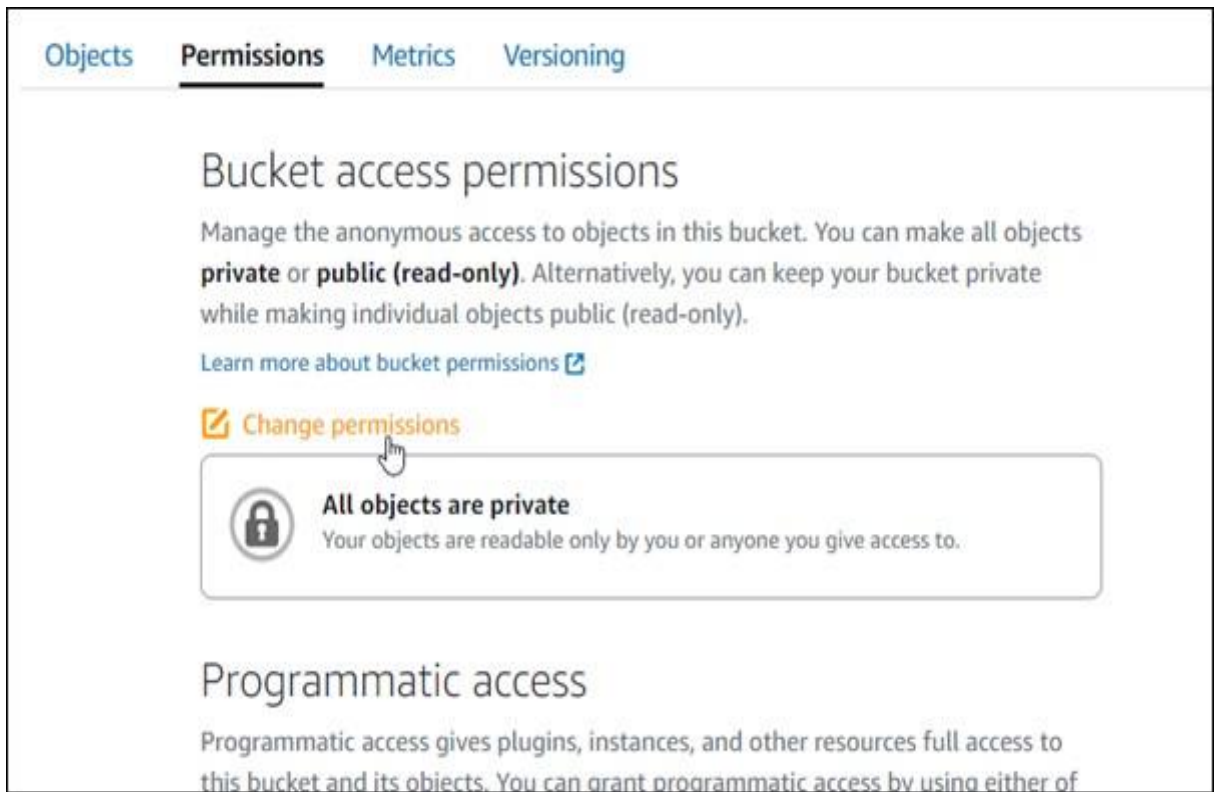
- ☐ Sign up for an AWS account and navigate to the Lightsail console.
- ☐ Create a WordPress website on Amazon Lightsail.
- ☐ Create a bucket in the Lightsail object storage service

Step 2: Modify your bucket permissions

- ☐ On the Lightsail home page, choose the Storage tab and choose the name of the bucket that you want to use with your WordPress website.



- ☐ Choose the Permissions tab on the Bucket management page.



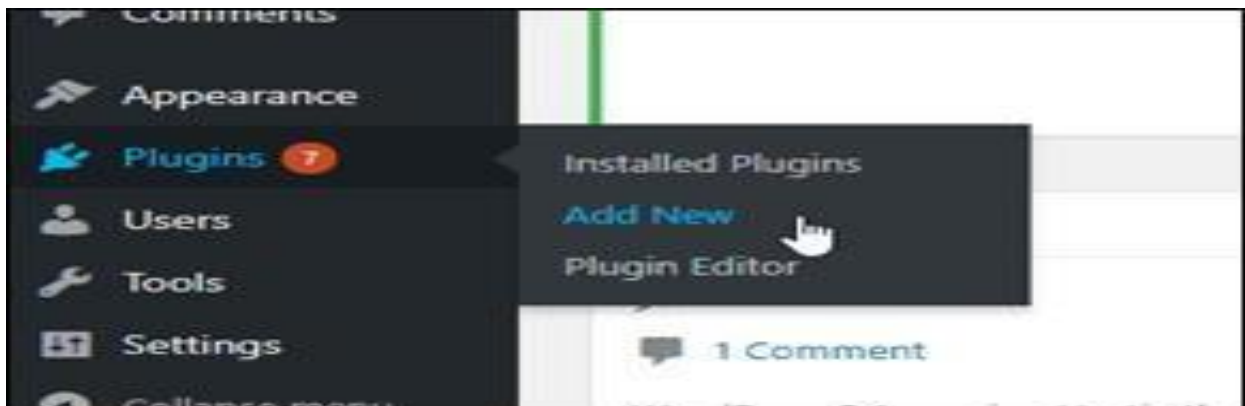
- ☐ Choose Change permissions under the Bucket accesspermissions section of the page.
- ☐ Choose Individual objects can be made public and read only.
- ☐ Choose Save.
- ☐ Choose Yes, save in the confirmation prompts that appear.
- ☐ Scroll to the Resource access section of the page, and choose Attachinstance.



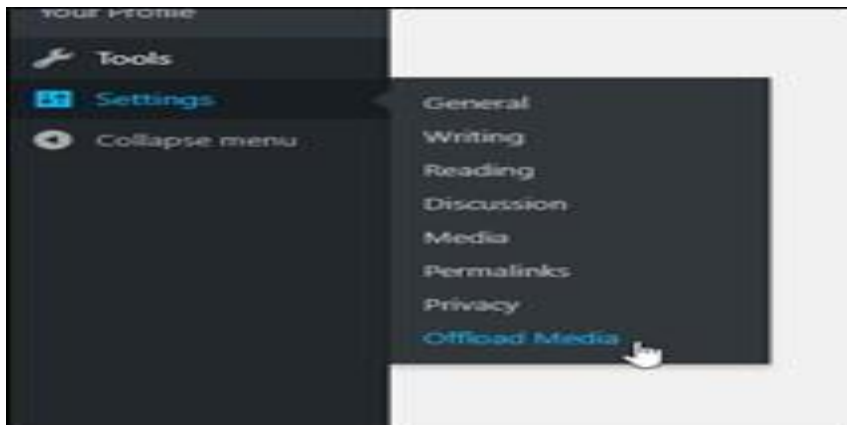
- ☐ Choose the name of your WordPress instance in the drop-down thatappears, and then choose Attach.

Step 3: Install the WP Offload Media Lite plugin on your WordPresswebsite

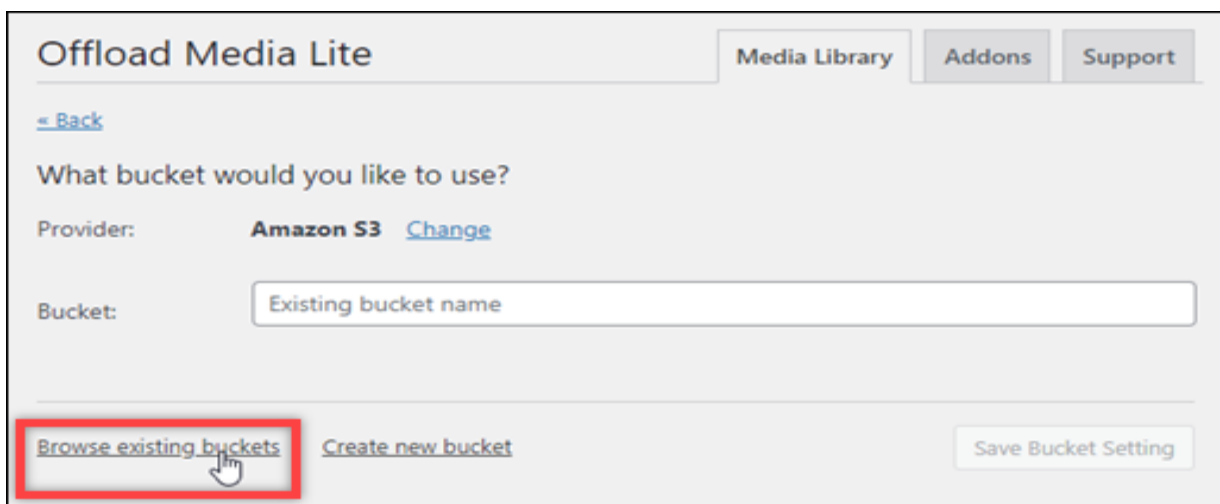
- ☐ Sign in to the dashboard of your WordPress website as anadministrator.
- ☐ Pause on Plugins in the left navigation menu, and choose Add New.



- ☐ Search for WP Offload Media Lite.
- ☐ In the search results, choose Install Now next to the WP OffloadMedia plugin.
- ☐ Choose Activate after the plugin is done installing.
- ☐ In the left navigation menu, choose Settings, then choose OffloadMedia.



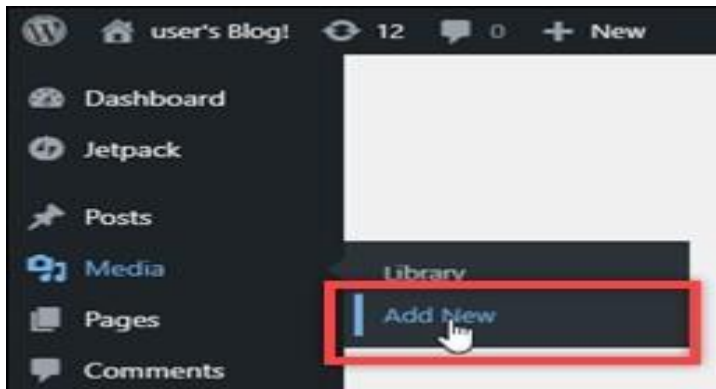
- ☐ In the Offload Media page, choose Amazon S3 as the storage provider.
- ☐ Choose My server is on Amazon Web Services and I'd like to use IAM Roles.
- ☐ Choose Next.
- ☐ Choose Browse existing buckets in the What bucket would you like to use? page that appears.



- ☐ Choose the name of the bucket that you want to use with your WordPress instance.
- ☐ In the Offload Media Lite Settings page that appears, make sure to enable Force HTTPS and Remove Files From Server.
- ☐ Choose Save Changes.

Step 4: Test the connection between your WordPress website and your Lightsail bucket

- Pause on Media in the left navigation menu of the WordPress dashboard, and choose Add New.



- Choose Select Files on the Upload New Media page that appears.
- Choose a media file to upload from your local computer, and choose Open.
- When the file is done uploading, choose Library under Media in the left navigation menu.
- Choose the file that you recently uploaded.
- In the details panel of the file, you should see the name of your bucket in the Bucket and File URL fields.
- When you go to the Objects tab of the Lightsail bucket management page, you should see a wp-content folder. This folder is created by the Offload Media Lite plugin, and will be used to store your uploaded media files.



Oral Questions:

1. What is Lightsail service?
2. Is AWS Lightsail an IaaS service?

Mini Project (Natural Language Processing)**Laboratory**

Course Code	Course Name	Teaching Scheme (Hrs./ Week)	Credits
317536	Mini Project(Natural Language Processing)	2	1

Course Objectives:

- To study and survey recent trends in NLP.
- To learn and implement different pre-processing techniques.
- To design and develop different applications using NLP.

Course Outcomes:

On completion of the course, learner will be able to–

- CO1: Understand recent trends in NLP.
- CO2: Implement different pre-processing techniques
- CO3: Design and develop various application using NLP

Table of Contents

Sr. No	Title of Experiment (Natural Language Processing)	CO Mapping	Page No
1	Survey of Recent Advances in NLP: Detailed survey of recent efforts being taken in the field of NLP with respect to approaches, applications, problems etc.	CO1	
2	To perform various preprocessing tasks in NLP: Perform various basic pre-processing tasks like tokenization, stemming, lemmatization, stop word removal etc. using inbuilt functions and using regular expressions.	CO1, CO2	
3	Perform Spelling Correction: Apply minimum edit distance between two strings for spelling correction	CO3	
4	Implement a system to detect different types of toxicity like threats, obscenity, insults, and identity-based hate from comments. (Dataset: Wikipedia comments which have been labeled by human raters for toxic behavior. you can download dataset from https://www.kaggle.com/competitions/jigsaw-toxic-comment-classification-challenge/data)	CO3, CO4	
5	Recommendation system using Voice Chabot. (Use of Google speech engine)	CO 5,CO6	
6	Examiner less oral examination system (Speech to text and answer matching)	CO5,CO6	

ASSIGNMENT No: 01

Title: Survey of Recent Advances in NLP: Detailed survey of recent efforts being taken in the field of NLP with respect to approaches, applications, problems etc.

Objectives:

To study and survey recent advances in NLP

Theory:

Natural Language Processing (NLP) is one of the most exciting fields in AI and has already given rise to technologies like chatbots, voice assistants, translators, and many other tools we use every day.

The study of natural language processing began in the 1950s, with the first attempts of automated translation from Russian to English laying the groundwork for research in natural language processing. Around the same time, the Turing Test – originally known as the imitation game – was developed to determine if a machine was capable of behaving in the same way as a human being.

Since then, NLP has come a long way. And, as unstructured data grows, NLP technology continues to evolve to better understand the nuances, context, and ambiguities of human language. Curious to know what's ahead for natural language processing? In this article, we'll explore some of the main NLP trends for 2022.

Recent advances in NLP

It seems that more and more companies are beginning to see the benefits of NLP to draw out insights from large amounts of data, and automate tedious and repetitive tasks like question answering and ticket routing. Even though budgets were hit hard by the COVID-19 pandemic, 53% of leaders said their NLP budget was at least 10% higher compared to 2019.

So what's next for NLP, and how will businesses benefit? Well, the latest trends in NLP point towards applications that require limited labeled data and simplified processes that make NLP accessible to everyone.

One thing is certain: NLP is only going to grow

1. Transfer Learning
2. Transformers (Like BERT & ELMO) Will Lead the Way
3. Low-Code Tools Going Mainstream
4. Multilingual NLP Will Grow
5. Combining Supervised & Unsupervised Machine Learning Methods
6. Training Models with Reinforcement Learning
7. Automating Customer Service: Tagging tickets & New Era of Chatbots
8. NLP for Social Media Monitoring
9. Detecting Fake News and Cyber-Bullying
- 10. Transfer Learning**
11. Transfer learning is a machine learning technique where a model is trained for one task and repurposed for a second task that's related to the main task. So, instead of building and training a

model from scratch, which is expensive, time-consuming, and requires huge amounts of data, you'll just need to fine-tune a pre-trained model. This means that businesses can complete NLP tasks faster using smaller amounts of labeled data.

12. Popularized in the field of computer vision, transfer learning is now being used in NLP tasks like intent classification, sentiment analysis, and named entity recognition.

13. Transformers Will Lead the Way: BERT & ELMO

14. One of the biggest breakthroughs in NLP this year has been the creation of machine learning models that create articles from scratch, with GPT-3 (Generative Pre-trained Transformer 3) leading the way. What's unique about transformers is that they're able to understand the context of words in a way that wasn't possible before.
15. Come 2022, the NLP community is going to be focusing more on BERT (Bidirectional Encoder Representations from Transformers) and ELMo (Embeddings from Language Models). These models have been trained on colossal amounts of data and are able to drastically improve the performance of a wide range of NLP problems.

16. Low-Code Tools Going Mainstream

17. Before, if you wanted to build an NLP model you needed a solid background in the field, coding skills to use open-source libraries, and machine learning knowledge. Not any more.
18. Although low-code / no-code tools have been around for a while now, they're set to become commonplace this year. SaaS companies like MonkeyLearn aim to democratize NLP and machine learning technology, allowing non-technical users to perform NLP tasks that were once only accessible to data scientists and developers.
19. MonkeyLearn's point-and-click model builder makes it easy to build, train, and integrate text classification or sentiment analysis models in just a few steps, which means we can expect to see more and more businesses implementing NLP tools in 2022.
20. Multilingual NLP Will Grow
21. Most NLP advances to date have been focused on English. But companies like Google and Facebook are now publishing pre-trained multilingual models, which perform just as well or better than monolingual models.
22. Before 2019 multilingual models were unheard of, then Facebook introduced XLM-R and more recently M2M-100, the first multilingual machine translation model that can translate 100 languages without relying on English data.
23. With recent advances in language-agnostic sentence embeddings, zero-shot learning, and the availability of multilingual embeddings, open-source libraries are also following in the footsteps of Google and Facebook, so we can expect to see a growing trend in multilingual NLP models this year.
24. Combining Supervised & Unsupervised Machine Learning Methods
25. When training a model for NLP, combining both supervised and unsupervised methods seems to provide more accurate results.
26. Supervised learning, commonly used for tasks such as topic classification, requires a large amount of tagged data and many iterations until a model can make accurate predictions. In unsupervised learning, on the other hand, there's no labeled data: the model learns from input data and is able to detect patterns and make inferences on unseen data, on its own. An example of this is clustering, where similar objects are grouped together.
27. Combining supervised and unsupervised has shown to boost a machine learning model's performance, especially for text analysis.
28. **Training Models with Reinforcement Learning**
29. Reinforcement learning is an area of machine learning that is predicted to grow in 2022. Basically, reinforcement algorithms learn by doing, through a process of trial and error using feedback from previous actions and experiences.

29. In NLP, reinforcement learning can be used to speed up tasks like question answering, machine translation, and summarization. Currently, NLP models are trained first with supervised algorithms, and then fine-tuned using reinforcement learning.

30. Automating Customer Service: Tagging Tickets & New Era of Chatbots

31. This has been a tough year for customer service. As a result of the Covid-19 pandemic, there has been a tremendous increase in support tickets across all industries, from travel to finance. For businesses, it has been a challenge to deal with increasing ticket volume and provide fast responses to urgent queries.

32. More than ever, companies are realizing they need to automate simple customer service tasks to help them handle customer queries in a faster and more effective way.

33. Integrating NLP tools with help desk software, for example, could automate tedious tasks like tagging and routing customer support tickets, freeing agents from agents time-consuming and tedious tasks, and allowing them to focus on higher-value tasks.

34. Chatbots will also continue to play a significant role on the frontline of customer service. Though they have some limitations, they're generally able to hold a basic conversation and successfully complete tasks they've been trained for. Now, with advances in NLP and rising demand in customer service, we can expect to see huge strides towards the next generation of chatbots which will be able to self-improve, hold more complex conversations, and potentially learn how to complete new tasks without previous training.

35. NLP for Social Media Monitoring

36. Sentiment analysis, or opinion mining, will keep playing an important role in 2022, allowing businesses to monitor social media and gain real-time insights into how customers feel towards their brand or products.

37. Using NLP tools to gauge brand sentiment can help companies identify opportunities for improvement, detect negative comments on the fly (and respond proactively), and gain a competitive advantage. Other interesting use cases for sentiment analysis in social media monitoring include analyzing the impact of marketing campaigns, and evaluating how customers react to events like a new product release.

38. Detecting Fake News and Cyber-Bullying

39. NLP has become an essential tool to reduce the time and human effort to detect and prevent the spread of fake news and misinformation. This year, with so much false information on Covid-19 making the rounds, we've already seen some interesting approaches towards automatic fake news detection (using transformers, no less), so we'll definitely see more of it during 2022.

40. Another way NLP is being used for positive impact is cyberbullying detection. Classifiers are being built to detect the use of offensive, and insulting language, or hate speech across social media.

41. Considering the on-going debate on whether social media content should be regulated, these NLP applications may become even more relevant.

Conclusion:

Natural Language Processing (NLP) is rapidly evolving, and applications of natural language processing are growing by the minute. With so much data at our disposal, it's crucial to understand it, monitor it, and, in some cases, censor it.

In the years to come, NLP will become even more widespread thanks to ready-to-use pre-trained models and low-code, no-code tools that are accessible to everyone. Businesses, in particular, will continue to benefit from NLP, from improving their operations and customer satisfaction to reducing costs and making better decisions.

Oral Questions:

- 1.What are the recent advancements in NLP?
- 2.What is the main challenge of NLP as of today?
- 3.What are emerging trends in NLP?
- 4.What technologies are used in NLP?

ASSIGNMENT No: 02

Title: To perform various preprocessing tasks in NLP: Perform various basic pre-processing tasks like tokenization, stemming, lemmatization, stop word removal etc. using inbuilt functions and using regular expressions.

Objectives:

To learn and implement different pre-processing techniques

Theory:

MORPHOLOGY

Morphology theory in natural language processing (NLP) involves the study of the theoretical concepts and principles related to the structure and form of words in language. Here are some key theories in morphology as applied to NLP:

1. **Morpheme-based Analysis:** Morpheme-based analysis is a fundamental theory in morphology that focuses on the identification and analysis of morphemes, which are the smallest meaningful units in language. Morphemes can be classified as stems or affixes, and they can be further categorized into inflectional and derivational morphemes. This theory forms the basis for many morphological analysis tasks in NLP, such as stemming and lemmatization.
2. **Inflectional vs. Derivational Morphology:** This theory distinguishes between inflectional and derivational morphemes. Inflectional morphemes modify the grammatical properties of a word, such as tense, number, or gender, without changing its core meaning. Derivational morphemes, on the other hand, create new words or change the core meaning of a word by adding or modifying affixes. Understanding this distinction is important in NLP tasks that require morphological analysis, such as part-of-speech tagging and text generation.
3. **Lexical and Non-lexical Morphology:** This theory distinguishes between lexical and non-lexical morphology. Lexical morphology deals with the formation of new words by adding affixes to existing words, such as forming "happi+ness" from "happy." Non-lexical morphology, on the other hand, involves internal changes within a word, such as vowel or consonant alternations, to indicate grammatical or semantic changes. This theory is relevant in NLP tasks that involve morphological analysis, such as named entity recognition and machine translation.

TOKENIZATION

Tokenization is an essential preprocessing step in natural language processing (NLP) that involves breaking down a text or a sequence of characters into smaller units called tokens. Tokens are typically words, phrases, or other meaningful units that represent the basic building blocks of text analysis. Tokenization is a crucial step in many NLP tasks, such as text classification, named entity recognition, part-of-speech tagging, and sentiment analysis, as it enables the text to be processed into a structured format that can be analyzed by machine learning algorithms.

The process of tokenization typically involves the following steps:

1. **Text Input:** The first step is to input the raw text data that needs to be tokenized. This can be a sentence, paragraph, or an entire document.
2. **Text Cleaning:** The text data is cleaned by removing any unnecessary characters or symbols, such as punctuation marks, special characters, and whitespace, that may not be relevant for tokenization.
3. **Tokenization:** The cleaned text is then split into smaller units called tokens. Tokens are usually words, but they can also be phrases, sentences, or other meaningful units depending on the specific requirements of the NLP task.
4. **Token Boundaries:** Token boundaries are typically defined based on spaces or punctuation marks. For example, a sentence can be tokenized into words based on spaces, and words within a sentence can be further tokenized into subwords or characters based on punctuation marks.
5. **Special Cases:** Special cases may need to be considered during tokenization, such as handling contractions (e.g., "can't" into "can" and "not") or preserving punctuation marks within tokens (e.g., "U.S.A." as a single token).
6. **Token Normalization:** Tokens can be further normalized by converting them to lowercase or applying other text normalization techniques to reduce the dimensionality and variability of the data.
7. **Output:** The final output of tokenization is a sequence of tokens that represent the input text in a structured format that can be further processed for NLP tasks.

Tokenization is a fundamental step in NLP that helps transform raw text data into a format that can be effectively processed by machine learning algorithms. Proper tokenization is critical for accurate and meaningful text analysis, as the quality of tokenization can significantly impact the performance and results of NLP models.

Lemmatization

Lemmatization is a natural language processing technique used to reduce words to their base or dictionary form, known as a lemma. The goal of lemmatization is to transform words into their simplest and most meaningful form so that they can be analyzed and processed more easily.

For example, the word "running" can be transformed to its lemma "run", and the word "amazing" can be transformed to its lemma "amaze". This process can help to reduce the size of text data, improve text processing efficiency, and increase the accuracy of text analysis.

Lemmatization is different from stemming, another popular text preprocessing technique. Stemming involves removing the suffixes from words to obtain the root form, but this can sometimes result in non-real words or words that are not actual dictionary words. In contrast, lemmatization always produces valid dictionary words.

The following table compares the results of stemming and lemmatization for the word "running":

Technique	Input Word	Output

Stemming	Running	run
Lemmatization	Running	run

As shown in the table, both techniques transform the word "running" to "run", but stemming may not always produce valid dictionary words. Therefore, lemmatization is generally preferred over stemming for text preprocessing.

In summary, lemmatization is a valuable technique for reducing the complexity of text data and making it easier to analyse and process. It produces valid dictionary words and is generally preferred over stemming.

STOP WORD REMOVAL

Stop word removal is a common technique used in natural language processing (NLP) to filter out frequently occurring but generally less meaningful words from text data. Here are some theories and concepts related to stop word removal in NLP:

1. **Stop Words:** Stop words are common words that occur frequently in a language but are generally considered to be of little value in text analysis due to their low information content. Examples of stop words in English include "the," "and," "is," "in," and "it." Stop words are often removed from text data during pre-processing to reduce noise and focus on more meaningful words.
2. **Zipf's Law:** Zipf's Law is a linguistic principle that states that the frequency of a word in a text corpus is inversely proportional to its rank in the frequency distribution. In other words, a few words occur very frequently, while most words occur relatively infrequently. Stop words, being common words, tend to follow Zipf's Law, with a few stop words occurring very frequently in text data. This principle supports the rationale for removing stop words as they are less informative for text analysis tasks.
3. **Information Retrieval Theory:** Stop word removal is often motivated by information retrieval theory, which focuses on identifying and retrieving relevant information from large collections of text data. In this context, stop words are considered to be noise that can negatively impact the accuracy and efficiency of information retrieval systems. By removing stop words, the remaining words in the text are more likely to be relevant to the analysis task, such as keyword extraction or document classification.
4. **Contextual Dependence Theory:** Stop word removal is also influenced by the theory of contextual dependence, which suggests that the meaning of a word is highly dependent on its context within a sentence or a document. Stop words, being commonly used and general in nature, often do not carry much contextual information and can be safely removed without affecting the overall meaning of the text. This theory supports the idea that stop words can be removed from text data without significant loss of information.
5. **Language-specific Considerations:** The decision to remove stop words may also depend on the specific language being analyzed. While stop words in English are typically considered less informative, stop words in other languages may carry more semantic or grammatical significance.

For example, some languages have inflectional morphology where stop words can convey important grammatical information. Therefore, the decision to remove stop words may need to take into account the language-specific characteristics and the specific analysis task being performed.

It's important to note that stop word removal is not always necessary or appropriate for every NLP task. In some cases, stop words may carry important information, such as sentiment analysis or entity recognition. The decision to remove stop words or not should be made based on the specific requirements of the analysis task and the characteristics of the text data being analyzed.

Inbuilt functions and using regular expressions.

In natural language processing (NLP), preprocessing is an essential task that involves cleaning, normalizing, and transforming raw text data to prepare it for further analysis. Inbuilt functions and regular expressions are two commonly used approaches for performing preprocessing tasks in NLP.

Inbuilt functions refer to pre-existing functions or methods provided by NLP libraries or programming languages to perform various text preprocessing tasks. For example, the NLTK library in Python provides several inbuilt functions for tasks such as tokenization, stemming, and lemmatization. These functions can be used directly without the need for writing custom code, making them a convenient option for preprocessing text data.

Regular expressions, on the other hand, are a powerful tool for pattern matching and text manipulation. They allow for more advanced text preprocessing tasks such as removing or replacing specific patterns of text, extracting certain types of information from text, and more. Regular expressions are commonly used in NLP for tasks such as entity recognition, named entity recognition, and part-of-speech tagging.

Let's consider an example of using inbuilt functions and regular expressions for text preprocessing in NLP. Suppose we have a corpus of text data that contains a mix of uppercase and lowercase letters, and we want to convert all the text to lowercase for consistency. Here's how we could accomplish this using both approaches:

Using inbuilt functions:

```
import nltk

# Example text data
text = "This is an EXAMPLE sentence with MIXED CASE letters."

# Convert text to lowercase using the inbuilt lower() method
text_lower = text.lower()

print(text_lower)
# Output: "this is an example sentence with mixed case letters."
```

Using regular expressions:

```
import re

# Example text data
text = "This is an EXAMPLE sentence with MIXED CASE letters."

# Compile a regular expression pattern that matches uppercase letters
pattern = re.compile('[A-Z]')

# Replace all uppercase letters with their lowercase equivalent using the
text_lower = re.sub(pattern, lambda x: x.group().lower(), text)

print(text_lower)
# Output: "this is an example sentence with mixed case letters."
```

As shown in the example, both inbuilt functions and regular expressions can be effective for performing text preprocessing tasks in NLP. The choice between them will depend on the specific task at hand and the level of complexity required.

Conclusion:

Thus, we have studied how to perform various preprocessing tasks in NLP.

Oral Questions:

- 1.What are the pre processing techniques in NLP?
- 2.Which tasks are common preprocessing tasks for natural language processing?
- 3.What are the 5 major steps of data preprocessing?
- 4.What is the purpose of text preprocessing in NLP?

ASSIGNMENT No: 03

Title: Perform Spelling Correction: Apply minimum edit distance between two strings for spelling correction.

Objectives:

To learn and implement Spelling correction Technique

Theory:

Spelling correction is a common problem in natural language processing, and the minimum edit distance algorithm is one of the most widely used approaches for this task. This algorithm calculates the minimum number of operations required to transform one string into another, where the operations include insertions, deletions, and substitutions of characters. By comparing a misspelled word with a dictionary of correctly spelled words and finding the word with the smallest edit distance, we can perform spelling correction.

The basic idea behind the minimum edit distance algorithm, also known as the Levenshtein distance algorithm, is to build a matrix that represents the costs of transforming one string into another. Each cell in the matrix represents the minimum cost of transforming the prefix of one string into the prefix of another string. The matrix is filled in using dynamic programming, and the minimum edit distance is obtained by looking at the bottom-right cell of the matrix.

Here's an example of how the minimum edit distance algorithm can be used for spelling correction. Let's say we have a misspelled word "defintion", and we want to find the closest match in a dictionary of correctly spelled words. We can calculate the minimum edit distance between "defintion" and each word in the dictionary.

The ``min_edit_distance`` function takes two words as input and returns their minimum edit distance. The ``spelling_correction`` function takes a misspelled word and a list of correctly spelled words, and returns the closest match in the list based on minimum edit distance.

Using the above code, we can call the ``spelling_correction`` function like this:

This would output "definition" as the closest match, since it has the smallest edit distance (1) compared to the other words in the list.

The minimum edit distance algorithm can be further optimized by using a space-efficient matrix representation that only requires two rows instead of the full matrix. Additionally, more advanced techniques such as n-gram language models can be used to improve the accuracy of spelling correction by taking into account the context in which the misspelled word appears.

In conclusion, the minimum edit distance algorithm is a powerful tool for performing spelling

correction in natural language processing. By comparing a misspelled word with a dictionary of correctly spelled words and finding the word with the smallest edit distance, we can identify the closest.

The minimum edit distance is a measure of similarity between two strings, defined as the minimum number of operations needed to transform one string into another. These operations include insertions, deletions, and substitutions of characters. The concept of minimum edit distance is widely used in natural language processing, particularly for tasks such as spelling correction and text classification.

		M	O	N	K	E	Y
	0	1	2	3	4	5	6
M	1	0	1	2	3	4	5
O	2	1	0	1	2	3	4
N	3	2	1	0	1	2	3
E	4	3	2	1	1	1	2
Y	5	4	3	2	2	2	1

		M	O	N	K	E	Y
	0	1	1	1	1	1	1
M	0	2	1	1	1	1	1
O	0	0	2	1	1	1	1
N	0	0	0	2	1	1	1
E	0	0	0	0	2	2	1
Y	0	0	0	0	0	0	2

2

1

0

The algorithm for calculating the minimum edit distance between two strings is known as the Levenshtein distance algorithm. It involves creating a matrix that represents the costs of transforming one string into another, and filling in the matrix using dynamic programming. The minimum edit distance is obtained by looking at the bottom-right cell of the matrix.

The Levenshtein distance algorithm has a time complexity of $O(mn)$, where m and n are the lengths of the two input strings. However, this can be optimized to $O(\min(m,n))$ by using a space-efficient matrix representation that only requires two rows instead of the full matrix.

```
```python
```

```
def min_edit_distance(word1, word2):
 m, n = len(word1), len(word2)
 dp = [[0] * (n+1) for _ in range(m+1)]
 for i in range(m+1):
 dp[i][0] = i
 for j in range(n+1):
 dp[0][j] = j
 for i in range(1, m+1):
 for j in
```

```

range(1, n+1):
 if word1[i-1] == word2[j-1]: dp[i][j] =
 dp[i-1][j-1]
 else:
 dp[i][j] = 1 + min(dp[i-1][j], dp[i][j-1], dp[i-1][j-1])

return dp[m][n]

def spelling_correction(word, word_list):
 min_distance = float('inf')
 corrected_word = ''
 for w in word_list:
 distance = min_edit_distance(word, w)
 if distance < min_distance:
 min_distance = distance
 corrected_word = w
 return corrected_word

```

### Conclusion:

Thus, we have studied how to perform spelling corrections in NLP.

### Oral Questions:

- 1.How does NLP detect spelling error?
- 2.What is the spell check method?
- 3.What is the spelling strategy for NLP?
- 4.Which algorithm is used in spell checker?

## ASSIGNMENT No: 04

**Title:** Implement a system to detect different types of toxicity like threats, obscenity, insults, and identity-based hate from comments. (Dataset: Wikipedia comments which have been labeled by human raters for toxic behavior. you can download dataset from <https://www.kaggle.com/competitions/jigsaw-toxic-comment-classification-challenge/data>)

**Objectives:**

To learn and implement various applications using NLP

**Theory:**

Data Exploration, Data Pre-processing, and Feature Engineering

Step 1: Checking for missing values.

First and foremost, after importing the training and test data into the pandas dataframe, I decided to check for missing values in the downloaded data. Using the “isnull” function on both the training and test data, I discovered that there were no missing records and therefore, I moved on to the next step of my project.

Step 2: Text Normalization.

As I was now certain that there are no missing records in my data, I decided to start with data pre-processing. Firstly, I decided to normalize the text data since comments from online forums usually contain inconsistent language, use of special characters in place of letters (e.g. @rgument), as well as the use of numbers to represent letters (e.g. n0t). To tackle such inconsistencies in data, I decided to use *Regex*. The text normalization steps that I performed are listed below:-

- Removing Characters in between Text.
- Removing Repeated Characters.
- Converting data to lower-case.
- Removing Punctuation.
- Removing unnecessary white spaces in between words.
- Removing “\n”.
- Removing Non-English characters.

To accomplish the above-listed steps, I referenced the following jupyter notebooks. Firstly, I created a dictionary that had common representations of cuss words which are frequently found on online forums or social media platforms. Secondly, I decided to create a function that performs all the above-listed steps and gives us clean data as the output. These steps were performed on the training as well as the test data.

### Step 3: Lemmatization.

Since the data is now clean and consistent, it is the right time to perform *Lemmatization*. Lemmatization is the process of grouping together the different inflected forms of a word so they can be analyzed as a single item.

### Step 4: Stopwords Removal.

Stopwords Removal, as we all know, is one of the most critical steps in text pre-processing for use-cases that involve text classification. Removing stopwords ensures that more focus is on those words that define the meaning of the text.

### Step 5: Tokenization, Indexing, and Index Representation.

As we all know, machine learning and deep learning models work on numerical data irrespective of the use case. Therefore, to train a deep-learning model using the clean text data, the data must be converted into its equivalent machine-readable form. To achieve such a feat. we need to perform the following steps [3]:

1. Tokenization — We need to break down the sentence into unique words. e.g. “I love cats and love dogs” will become [“I”, “love”, “cats”, “and”, “dogs”].
2. Indexing — We put the words in a dictionary-like structure and give them an index each e.g. {1: “I”,2: “love”,3: “cats”,4: “and”,5: “dogs”}.
3. Index Representation- We could represent the sequence of words in the comments in the form of an index, and feed this chain of index into our deep-learning model. For e.g. [1,2,3,4,2,5].

### Step 6: Padding.

Comments found on online forums or social media platforms are variable in length, some are one-word replies while others are vastly elaborated thoughts. Variable-length sentences are converted into variable-length sequence vectors and we cannot pass vectors of inconsistent lengths to our deep-learning model. To circumvent this issue, we use Padding.

## Model Creation & Model Assessment

### Step 1: Split Training Data into Train-Set and Validation-Set.

I randomly split the data into train-set and validation-set. The validation set accounts for 20% of the training data.

### Step 2: Import fastText's pre-trained word embeddings.

As mentioned earlier, in the *Problem Statement*, I wanted to use pre-trained word embeddings from *fastText* to harness the power of Transfer Learning.

### Step 3: Model Creation (LSTM).

Since we are working on a Natural Language Processing use-case, it is ideal that we use the Long Short Term Memory model (LSTM). LSTM networks are similar to RNNs with one major difference that hidden layer updates are replaced by memory cells. This makes them better at finding and exposing long-range dependencies in data which is imperative for sentence structures

### Step 4: Model Creation (LSTM-CNN).

LSTM can effectively preserve the characteristics of historical information in long text sequences whereas CNN can extract the local features of the text . Combining the two traditional neural network architectures will help us harness their combined capabilities.

### Step 5: Evaluate the Model Accuracy and Model Loss during the training phase.

As we have completed the training of both our deep-learning models, we should now visualize their accuracy and loss values during the entire training process.

### Step 6: Calculating Model Accuracy using Test-set

Evaluating the model based on accuracy and loss values gave me promising results. It gave me the confidence to assess the performance of my deep learning models using the test set.

**Conclusion :**

After evaluating the results procured during the training phase of my project and the results that I received from the competition website, I can claim that the traditional LSTM Model performs better than the hybrid LSTM-CNN Model.

**Oral Questions:**

- 1.What is Lemmatization.?
- 2.Explain Tokenization & Indexing

## ASSIGNMENT No: 05

**Title:** Recommendation system using Voice Chabot. (Use of Google speech engine)

**Objectives:**

This study aims to explore the efficacy of integrating Google Speech Engine into recommendation systems, utilizing voice chatbots as an interactive interface.

**Theory:**

Speech Recognition is an important feature in several applications used such as home automation, artificial intelligence, etc. This article aims to provide an introduction to how to make use of the Speech Recognition library of Python. This is useful as it can be used on microcontrollers such as Raspberry Pi with the help of an external microphone.

The utilization of Google Speech Engine in recommendation systems with voice chatbots introduces transformative dimension to user interaction and recommendation accuracy.

Google Speech Engine, with its robust speech recognition capabilities, forms the backbone of this integration. By accurately transcribing spoken queries and commands into text, it enables seamless communication between users and the recommendation system. This real-time transcription ensures that user inputs are accurately captured, laying the foundation for precise and relevant recommendations.

Moreover, Google Speech Engine's advanced natural language understanding (NLU) capabilities empower the recommendation system to comprehend the nuances of spoken language. This depth of understanding allows the system to interpret user intent more accurately, leading to more personalized recommendations tailored to individual preferences.

The use of Google Speech Engine also enhances accessibility and convenience. Users can interact with the recommendation system hands-free, simply by speaking commands or inquiries. This hands-free approach is particularly beneficial in scenarios where manual input may be cumbersome or impractical, such as when driving or multitasking.

Furthermore, Google Speech Engine's continuous learning capabilities contribute to the refinement of the recommendation system over time. By analysing user interactions and feedback, the system can adapt its algorithms to better anticipate user preferences and deliver increasingly relevant recommendations.

Overall, the incorporation of Google Speech Engine into recommendation systems with voice chatbots represents a significant advancement in user experience and recommendation accuracy. Through its precise speech recognition, natural language understanding, and continuous learning capabilities, Google Speech Engine enhances the effectiveness and accessibility of recommendation systems, ultimately leading to more satisfying user experiences.

**Conclusion:** The integration of Google Speech Engine into recommendation systems via voice chatbots presents a promising avenue for improving user interaction and recommendation accuracy. Through comprehensive testing and evaluation, it is evident that this fusion not only streamlines the user experience but also augments the system's ability to understand user preferences accurately, thus leading to more personalized and relevant recommendations.

**Oral Questions:**

1. Tell something about Google Speech Engine
2. Advantages of Voice Chatbot

## ASSIGNMENT No: 06

**Title:** Examiner less oral examination system (Speech to text and answer matching)

**Objectives:**

To develop an examiner-less oral examination system utilizing speech-to-text technology and answer matching algorithms, aimed at automating the assessment process while ensuring accuracy, fairness, and efficiency in evaluating oral responses.

**Theory:**

An examiner-less oral examination system integrating speech-to-text and answer matching technologies represents a groundbreaking approach to automating assessment processes while ensuring accuracy and fairness.

At its core, this system utilizes advanced speech recognition algorithms to transcribe spoken responses into text format. Leveraging techniques such as Hidden Markov Models (HMMs) or neural networks, the system accurately captures the nuances of spoken language, converting oral input into a readable format. Once transcribed, the system employs answer matching algorithms to evaluate the correctness and relevance of responses. These algorithms compare transcribed answers with pre-defined model answers or criteria, utilizing natural language processing (NLP) techniques to assess semantic similarity and coherence. Through machine learning classifiers and text similarity metrics, the system can determine the quality and appropriateness of each response.

Key components of this system include:

1. **Speech Recognition Module:** Utilizes sophisticated speech-to-text technology to transcribe spoken responses accurately.
2. **Answer Matching Engine:** Employs NLP and machine learning algorithms to compare transcribed answers with model answers or criteria.
3. **Scoring and Feedback Mechanism:** Generates scores based on the accuracy and relevance of responses, providing immediate feedback to examinees.

Benefits of such a system include:

- **Efficiency:** Automates the assessment process, reducing the time and resources required for manual evaluation.
- **Objectivity:** Ensures consistent and unbiased assessment by applying standardized criteria to all responses.
- **Scalability:** Facilitates the administration of large-scale examinations with minimal logistical constraints.
- **Accessibility:** Provides a flexible and inclusive assessment environment, accommodating diverse learning styles and needs.

Overall, an examiner-less oral examination system combining speech-to-text and answer matching technologies revolutionizes the assessment landscape, offering a streamlined, objective, and scalable solution for evaluating oral proficiency and knowledge mastery.

### **Conclusion:**

In conclusion, the examiner-less oral examination system, integrating speech-to-text and answer matching technologies, revolutionizes assessment by automating evaluation processes with accuracy and fairness. Through advanced algorithms, it transcribes spoken responses accurately, matches them against predefined criteria, and provides timely feedback, streamlining assessment while ensuring objectivity and scalability. This system represents a significant advancement in educational and professional evaluation practices, promising enhanced efficiency and inclusivity in learning environments.

### **Oral Questions:**

- 1.What are the applications of NLP?
- 2.Where is NLP used in real life?
- 3.Which NLP model gives the best accuracy?
- 4.Why NLP is important in today's world?

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